

Retinal Disease Classification and Severity Estimation from OCT Images using Ensemble Learning

A
MAJOR PROJECT REPORT

**Submitted in the partial Fulfillment of the requirements for the award of the
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**BACHELOR OF TECHNOLOGY
IN
ELECTRONICS AND COMMUNICATION ENGINEERING**

Submitted By

1. D. Chinmayee	22R21A0485
2. M. Sravya Sri	22R21A04A1
3. Y. Sri Vidya	22R21A04C8

UNDER THE GUIDANCE OF

Dr. V. Vijay Kumar
(Assistant Professor)



MLR Institute of Technology

(Autonomous)

(Affiliated to JNTUH, Hyderabad)

Dundigal, Hyderabad-500043

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Internal Guide

Dr.V. Vijay Kumar

Head of the Department

Dr. S.V.S. PRASAD

External Examiner

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Project associates:

D. Chinmayee	22R21A0485
M. Sravya Sri	22R21A04A1
Y. Sri Vidya	22R21A04C8

ABSTRACT

Retinal diseases such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen are major causes of vision impairment, requiring early and accurate diagnosis for effective treatment. This project presents an end-to-end deep learning-based system for automated retinal disease classification and severity estimation using Optical Coherence Tomography (OCT) images. A subset of four clinically significant classes (CNV, DME, DRUSEN, and NORMAL) is selected from the Retinal OCT (C8) dataset to ensure focused and reliable classification. The proposed system integrates advanced preprocessing, medically safe data augmentation, and multiple transfer learning models including DenseNet121, EfficientNetB4, and ResNet50V2, which are combined using a weighted ensemble approach to improve prediction accuracy and robustness. To enhance interpretability, Grad-CAM is utilized to highlight important retinal regions influencing model decisions. Additionally, a multi-stage clinical decision framework is developed to perform disease detection, classification, severity estimation, and reliability assessment. The primary use of this system is to assist ophthalmologists by providing fast, accurate, and interpretable diagnostic support, thereby reducing manual workload and enabling early detection of retinal diseases in real-world clinical settings.

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CHAPTER-1

INTRODUCTION

1.1 Retinal Diseases and Their Clinical Impact:

Retinal diseases are among the leading causes of vision impairment and blindness worldwide, affecting millions of individuals across different age groups. The retina is a vital component of the human eye responsible for capturing light and converting it into neural signals for visual perception. Any structural abnormality in the retina can significantly degrade vision quality and, in severe cases, result in permanent vision loss. Common retinal diseases include Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen. CNV is characterized by the growth of abnormal blood vessels beneath the retina, causing leakage and distortion of vision. DME occurs due to fluid accumulation in the retinal layers, typically associated with diabetic retinopathy. Drusen refers to the presence of yellow deposits under the retina and is considered an early indicator of age-related macular degeneration. In contrast, a normal retina exhibits well- defined and uniform retinal layers without abnormalities.

Early detection and timely treatment of these conditions are essential to prevent irreversible damage. However, conventional diagnosis relies on manual examination of retinal images by ophthalmologists, which is time-consuming and subject to inter-observer variability. With the increasing number of patients and the rapid growth of medical imaging data, there is a strong demand for automated and reliable diagnostic systems. Retinal diseases are among the leading causes of vision impairment and blindness worldwide, affecting millions of individuals across different age groups. The retina is a vital component of the human eye responsible for capturing light and converting it into neural signals for visual perception. It consists of multiple layers that work together to process visual information. Any structural or functional abnormality in the retina can significantly degrade vision quality and, in severe cases, result in permanent vision loss.

Common retinal diseases include Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen. CNV is characterized by the abnormal growth of new blood vessels beneath the retina, which can leak fluid or blood and cause severe distortion of central vision. DME occurs due to the accumulation of fluid in the macular region of the retina, typically associated with diabetic retinopathy, leading to blurred or impaired vision. Drusen refers to the presence of yellow deposits under the retina and is considered an early indicator of age-related macular degeneration. In contrast, a normal retina exhibits well- defined and

uniform retinal layers without any structural abnormalities.

The impact of retinal diseases extends beyond vision loss, affecting the quality of life, independence, and productivity of patients. Early symptoms may often go unnoticed, which makes timely diagnosis critical. If not detected at an early stage, these diseases can progress rapidly and lead to irreversible damage. Therefore, regular screening and accurate diagnosis play a crucial role in effective treatment and management.

Traditional diagnostic methods involve manual examination of retinal images by ophthalmologists using specialized imaging tools. While these methods are effective, they are time-consuming and depend heavily on the expertise of clinicians. Additionally, inter-observer variability can lead to inconsistencies in diagnosis. With the increasing volume of medical imaging data and the growing number of patients, manual analysis alone is insufficient to meet clinical demands.

In this context, Optical Coherence Tomography (OCT) has emerged as a widely adopted non-invasive imaging technique for retinal analysis. It provides high-resolution cross-sectional images of the retina, enabling detailed visualization of its internal structure. OCT plays a crucial role in detecting retinal abnormalities, monitoring disease progression, and evaluating treatment outcomes. However, interpreting OCT images requires expertise and can still be challenging when dealing with large datasets.

To address these challenges, there is a growing need for automated and intelligent diagnostic systems that can assist clinicians in analyzing OCT images efficiently. The integration of advanced technologies such as deep learning has shown significant potential in improving the accuracy, speed, and consistency of retinal disease diagnosis. Such systems can support ophthalmologists in early detection and decision-making, ultimately contributing to better patient outcomes.

1.2 Optical Coherence Tomography (OCT) :

Optical Coherence Tomography (OCT) is a non-invasive imaging modality widely used in ophthalmology for high-resolution visualization of retinal structures. It utilizes low-coherence light to capture cross-sectional images of biological tissues, particularly the retina, enabling detailed analysis of its internal layers. OCT has significantly advanced retinal diagnostics by providing micrometer-scale resolution, making it comparable to histological imaging without the need for tissue extraction.

Over the past decades, OCT has become an essential diagnostic tool in clinical ophthalmology due to its ability to detect early-stage retinal abnormalities. It plays a crucial role in diagnosing conditions such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen, which are major contributors to vision loss. The technique is fast, safe, and does not involve exposure to ionizing radiation, making it suitable for repeated use in monitoring disease progression.

The increasing availability of OCT imaging has resulted in large volumes of retinal data, creating challenges for manual interpretation. This has led to the integration of computational techniques, particularly deep learning, to assist in automated analysis. The combination of OCT imaging and artificial intelligence has the potential to significantly improve diagnostic accuracy, efficiency, and consistency in clinical practice.

1.2.1 Overview:

Optical Coherence Tomography (OCT) is an advanced imaging technique that provides detailed cross-sectional visualization of retinal structures with high resolution. It is widely used in ophthalmology due to its ability to generate precise images of retinal layers, enabling accurate diagnosis and monitoring of various eye diseases. OCT operates by measuring the intensity of backscattered light from different tissue layers, producing images that resemble histological sections of the retina.

One of the key advantages of OCT is its ability to provide non-invasive and real-time imaging. Unlike traditional diagnostic methods, OCT does not require direct contact with the eye or exposure to harmful radiation. This makes it safe for repeated use, allowing clinicians to monitor disease progression over time. The high-speed image acquisition capability of modern OCT systems further enhances its usability in clinical settings.

OCT images are typically represented as grayscale cross-sectional scans, where variations in intensity correspond to differences in tissue reflectivity. Healthy retinal layers appear as well-organized and continuous structures, while diseased conditions exhibit abnormalities such as fluid accumulation, layer disruption, or abnormal growth patterns. These structural differences play a crucial role in distinguishing between normal and pathological conditions.

The below Figure 1 shows a cross-sectional OCT image of the retina, displaying its different structural layers.

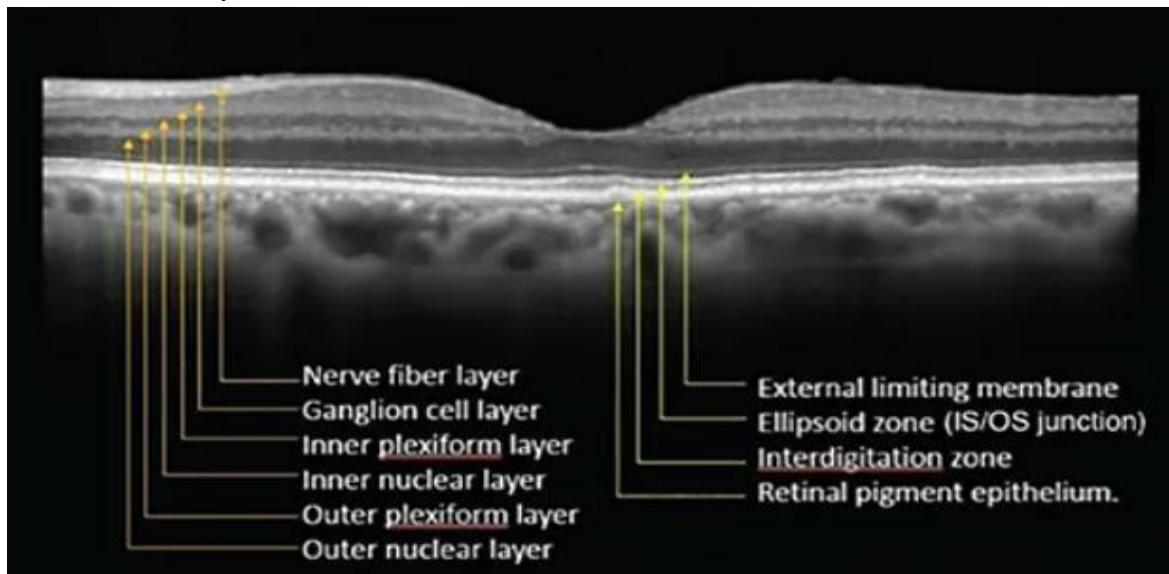


Figure 1: Cross-sectional OCT image showing different retinal layers

With technological advancements, OCT systems have evolved significantly. Early time-domain OCT systems have been replaced by more advanced spectral-domain (SD-OCT) and swept-source (SS-OCT) systems, which offer higher resolution, faster scanning speeds, and improved image quality. These advancements have enabled better visualization of deeper retinal layers and enhanced diagnostic accuracy. In addition to structural imaging, OCT has been extended to include functional imaging techniques such as OCT angiography (OCTA), which allows visualization of blood flow within retinal vessels without the need for contrast agents. This capability is particularly useful in detecting vascular abnormalities associated with retinal diseases.

The widespread adoption of OCT in clinical practice has resulted in the generation of large volumes of imaging data. While this provides valuable insights into retinal health, it also poses challenges for manual interpretation. Analyzing these images requires expertise and can be time-consuming, especially when dealing with large datasets.

To overcome these challenges, automated analysis techniques based on artificial intelligence and deep learning are increasingly being integrated with OCT imaging. These techniques enable efficient feature extraction and classification of retinal diseases, reducing the burden on clinicians and improving diagnostic consistency. As a result, OCT combined with intelligent systems is emerging as a powerful tool for modern healthcare applications.

Overall, Optical Coherence Tomography has transformed retinal imaging by providing detailed, fast, and reliable visualization of retinal structures. Its ability to detect subtle changes in tissue morphology makes it an indispensable tool for early diagnosis, disease monitoring, and treatment evaluation in ophthalmology.

1.2.2 Working Principle:

The working principle of Optical Coherence Tomography (OCT) is based on the concept of low-coherence interferometry, which enables high-resolution imaging of biological tissues. OCT uses light waves to capture depth-resolved information by measuring the time delay and intensity of light reflected from different layers of the retina. This allows the system to generate detailed cross-sectional images of retinal structures.

An OCT system primarily consists of three main components: a broadband light source, an interferometer, and a detection system. The emitted light beam is divided into two paths using a beam splitter: a reference beam and a sample beam. The sample beam is directed towards the retina, where it penetrates the tissue and is partially reflected by various retinal layers. Each layer reflects light differently based on its optical properties.

The reference beam, on the other hand, travels a known path length and is reflected back by a reference mirror. When the reflected sample beam and reference beam recombine, they produce an interference pattern. This interference occurs only when the optical path lengths of the two beams match within the coherence length of the light source. By analyzing this interference signal, the system determines the depth and intensity of reflections from different layers of the retina.

This Figure 2 illustrates the working principle of Optical Coherence Tomography using interferometry.

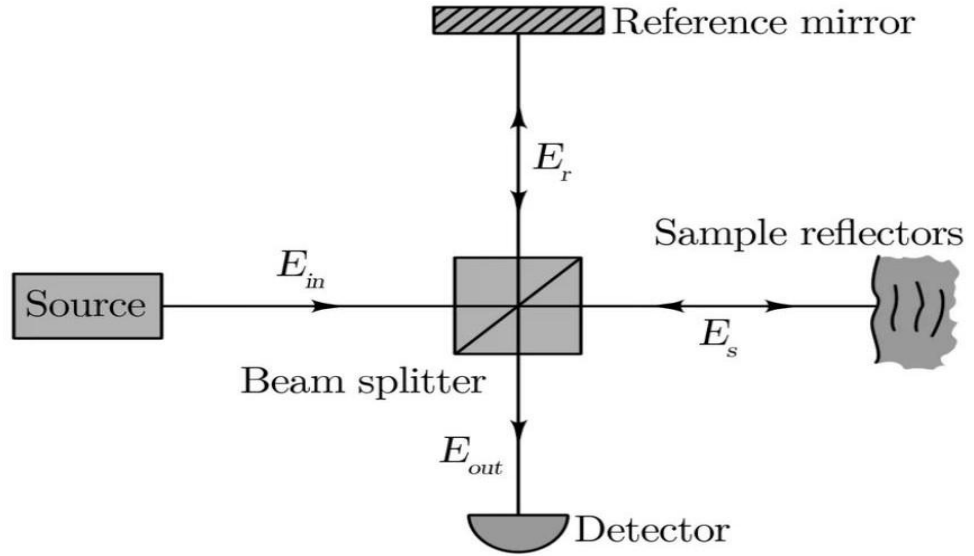


Figure 2: Working principle of OCT using Interferometry

The OCT system performs rapid scanning across the retinal surface to collect multiple depth profiles, also known as A-scans. These A-scans are combined to form two-dimensional cross-sectional images called B-scans. By stacking multiple B-scans, a three-dimensional representation of the retina can be generated, providing comprehensive structural information.

The resolution of OCT imaging depends on the coherence length of the light source, while the imaging depth depends on the wavelength of the light used. Shorter wavelengths provide higher resolution images, whereas longer wavelengths allow deeper penetration into tissue. Modern OCT systems optimize these parameters to achieve a balance between image clarity and depth.

Advanced OCT techniques such as Spectral-Domain OCT (SD-OCT) and Swept-Source OCT (SS-OCT) have significantly improved imaging speed and resolution. These systems use Fourier domain analysis to capture depth information more efficiently, enabling real-time imaging and enhanced visualization of retinal structures.

Overall, the working principle of OCT allows for precise, non-invasive, and high-speed imaging of the retina. This capability is essential for detecting structural abnormalities such as fluid accumulation, layer disruption, and abnormal tissue growth, which are key indicators of retinal diseases.

1.2.3 Applications in Ophthalmology:

Optical Coherence Tomography (OCT) is widely used in ophthalmology for the diagnosis, monitoring, and management of various retinal diseases. Its ability to provide high-resolution cross-sectional images of retinal layers makes it an essential tool in modern clinical practice. OCT is extensively used for the detection of retinal diseases such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen. In CNV, OCT helps identify abnormal blood vessel growth beneath the retina along with fluid leakage and structural distortions. In DME, it is used to detect fluid accumulation and retinal thickening, which are key indicators of the disease. For Drusen, OCT enables visualization of deposits beneath the retina, which are early signs of age-related macular degeneration.

The given Figure 3 below presents OCT images representing different retinal conditions such as Drusen, CNV, DME, and Normal.

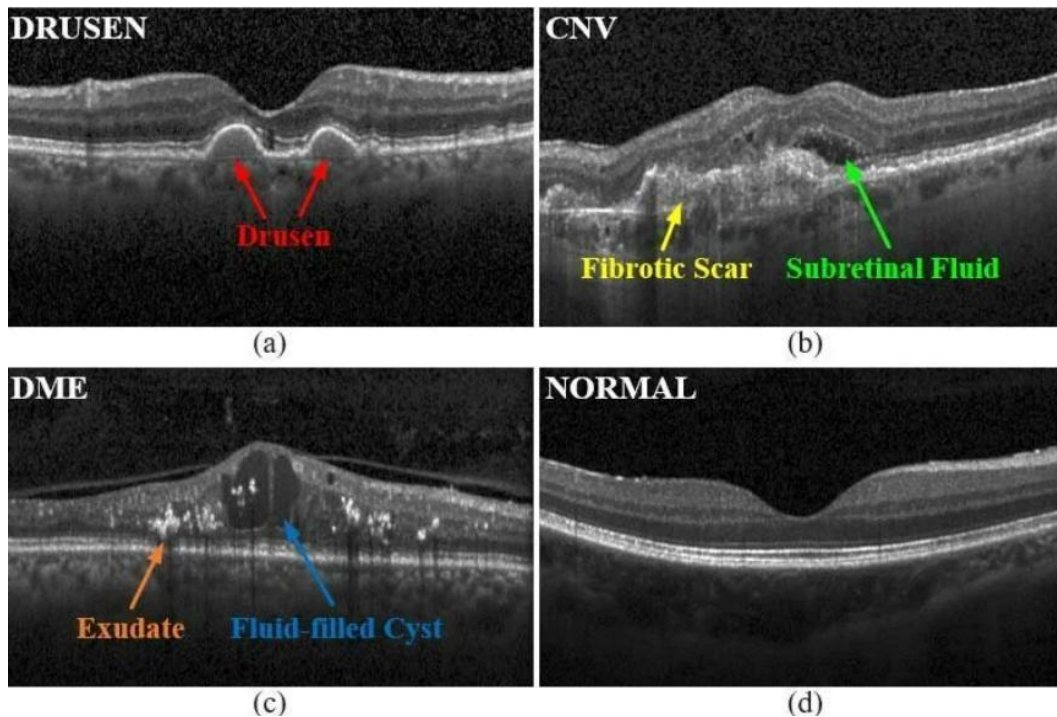


Figure 3: OCT images showing different retinal conditions: (a) Drusen, (b) CNV, (c) DME, and (d) Normal retina

In addition to disease detection, OCT plays an important role in monitoring disease progression. By comparing sequential scans over time, clinicians can observe structural changes in the retina and assess whether the condition is improving or worsening. This is particularly useful in chronic diseases where continuous monitoring is required. OCT is also used to evaluate treatment effectiveness. It helps clinicians analyze how well a patient is responding to

therapies such as anti-VEGF injections by observing changes in retinal thickness and fluid levels. Based on these observations, treatment plans can be adjusted accordingly.

Furthermore, OCT is applied in the analysis of the optic nerve and retinal nerve fiber layer, which is important for the early detection of glaucoma. Its non-invasive nature, high accuracy, and ability to provide detailed structural information make OCT a reliable tool for routine clinical use.

Overall, OCT has become an indispensable imaging technique in ophthalmology, enabling accurate diagnosis, effective monitoring, and improved clinical decision-making in the management of retinal diseases.

1.3 Deep Learning in Medical Image Analysis:

Deep learning has emerged as a powerful technique in the field of medical image analysis, enabling automated detection, classification, and interpretation of complex medical data. It is a subset of artificial intelligence that utilizes neural networks with multiple layers to learn hierarchical representations of data. In recent years, deep learning has significantly improved the accuracy and efficiency of medical image diagnosis, particularly in areas such as radiology, pathology, and ophthalmology.

In medical imaging, deep learning models are capable of extracting meaningful features directly from raw images without the need for manual feature engineering. This makes them highly effective for analyzing complex patterns and structures present in medical images such as OCT scans. The ability to process large volumes of data and identify subtle variations has made deep learning an essential tool for modern healthcare applications.

1.3.1 Working of CNN Models:

Convolutional Neural Networks (CNNs) are a class of deep learning models specifically designed for image processing and analysis. They are widely used in medical image analysis due to their ability to automatically learn spatial features from images. Unlike traditional machine learning methods, CNNs do not require manual feature extraction, as they can learn relevant patterns directly from raw image data.

A typical CNN architecture consists of multiple layers, including convolutional layers, pooling layers, and fully connected layers. The convolutional layer is the core component of a CNN, where filters (kernels) are applied to the input image to detect important features such as

edges, textures, and patterns. These filters slide across the image and generate feature maps that highlight different aspects of the input.

Following the convolutional layers, pooling layers are used to reduce the spatial dimensions of the feature maps. This helps in decreasing computational complexity while retaining the most important features. Common pooling operations include max pooling and average pooling, which summarize the information within a region of the feature map.

As the data passes through deeper layers of the network, the model learns more complex and abstract features. Initial layers focus on basic features like edges, while deeper layers capture high-level patterns such as shapes and structures. These learned features are then passed to fully connected layers, which perform the final classification based on the extracted information.

In medical image analysis, CNNs are highly effective for detecting subtle variations in images that may not be easily visible to the human eye. For retinal OCT images, CNNs can learn to distinguish between normal and diseased retinal structures by analyzing patterns such as fluid accumulation, layer disruption, and abnormal tissue formations.

The training process of a CNN involves feeding labeled images into the network, computing predictions, and adjusting the model parameters using optimization algorithms such as backpropagation and gradient descent. Over time, the model improves its ability to accurately classify images based on learned features.

Overall, CNNs provide a powerful framework for automated image analysis, enabling accurate and efficient classification of medical images. Their ability to learn hierarchical feature representations makes them particularly suitable for applications such as retinal disease detection using OCT images.

The Figure 4 below shows the diagram of the architecture of a Convolutional Neural Network (CNN).

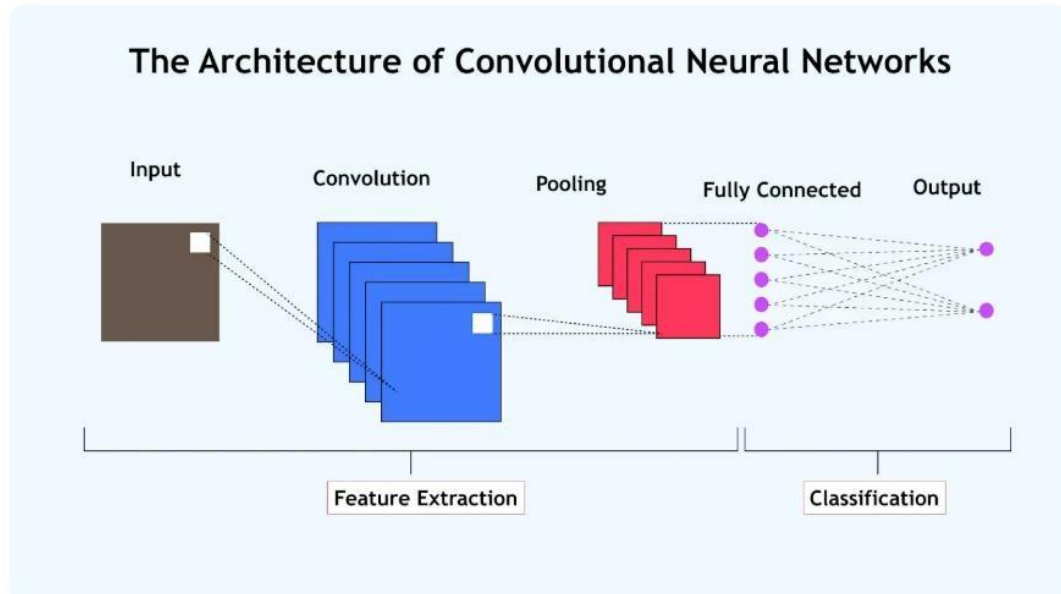


Figure 4: Architecture of a Convolutional Neural Network (CNN)

1.3.2 Role of Transfer Learning:

Transfer learning is a widely used technique in deep learning that enables the reuse of knowledge from pretrained models for new tasks. Instead of training a model from scratch, transfer learning utilizes models that have already been trained on large-scale datasets such as ImageNet and adapts them to specific applications. This approach significantly reduces training time and improves model performance.

In medical image analysis, collecting large labeled datasets is often challenging. Transfer learning addresses this limitation by leveraging feature representations learned from general images and applying them to domain-specific tasks such as retinal disease classification. The early layers of pretrained models capture basic features such as edges and textures, while deeper layers learn more complex patterns. These learned features can be fine-tuned using OCT images to achieve accurate classification results.

In this project, pretrained architectures such as DenseNet121, EfficientNet, and ResNet are employed. These models are initially trained on large datasets and then fine-tuned using retinal OCT images to adapt to the specific task of disease classification. This approach improves accuracy, reduces overfitting, and enhances generalization performance.

Transfer learning also enables efficient use of computational resources, as it requires fewer training epochs compared to training a model from scratch. Additionally, it allows the integration of advanced architectures without the need for extensive data collection.

The Figure 5 demonstrates transfer learning using feature extraction and fine-tuning.

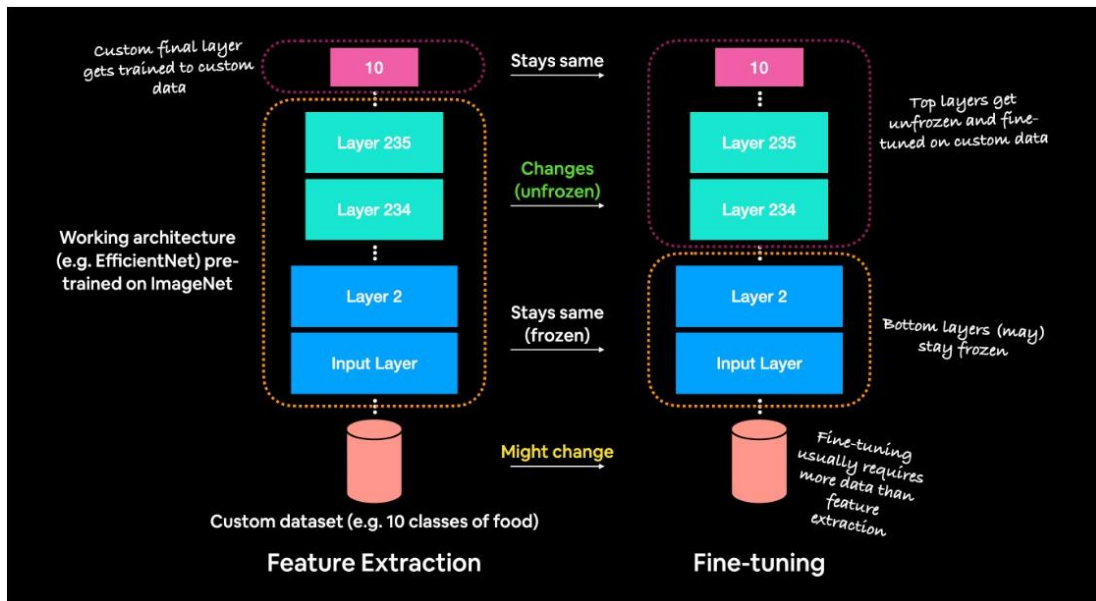


Figure 5: Transfer learning using feature extraction and fine-tuning

Overall, transfer learning plays a crucial role in modern medical image analysis by enabling high-performance models with limited data. Its ability to leverage pretrained knowledge makes it an essential component in developing robust and efficient deep learning systems for retinal disease detection.

1.3.3 Applications in Healthcare:

Deep learning has a wide range of applications in healthcare, particularly in medical image analysis, where it enables automated and accurate diagnosis of diseases. Its ability to learn complex patterns from large datasets makes it highly effective in analyzing medical images such as X-rays, MRI scans, CT scans, and OCT images.

In ophthalmology, deep learning models are widely used for the detection and classification of retinal diseases. By analyzing OCT images, these models can identify abnormalities such as fluid accumulation, layer distortion, and structural changes in the retina. This enables early detection of conditions such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen.

Deep learning also plays a significant role in improving diagnostic efficiency by reducing reliance on manual analysis. Automated systems can process large volumes of medical data quickly and provide consistent results, thereby minimizing human errors and inter-observer variability. This is particularly beneficial in clinical environments with a high patient load.

Another important application is the use of explainable artificial intelligence techniques such as Grad-CAM, which highlight important regions in medical images that influence the model's predictions. This improves transparency and enhances clinicians' trust in automated systems. Furthermore, deep learning is applied in disease progression analysis and treatment planning. By analyzing sequential medical images, models can track changes over time and assist clinicians in evaluating the effectiveness of treatments.

Overall, deep learning is transforming healthcare by enabling intelligent, automated, and reliable diagnostic systems, thereby improving both accuracy and efficiency in medical decision-making.

1.4 Problem Statement:

Retinal diseases such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen are among the leading causes of vision impairment and blindness worldwide. Early detection and accurate diagnosis of these conditions are essential to prevent irreversible vision loss. Optical Coherence Tomography (OCT) is widely used for retinal imaging due to its ability to provide high-resolution cross-sectional images of retinal layers. However, the interpretation of OCT images is complex and requires significant clinical expertise.

Manual analysis of OCT scans is time-consuming and becomes increasingly challenging with the growing volume of medical imaging data. In addition, variations in interpretation among clinicians may lead to inconsistencies in diagnosis. The subtle differences between normal and diseased retinal structures further increase the difficulty of accurate classification.

Therefore, there is a need to address the challenges associated with efficient, accurate, and consistent analysis of OCT images for reliable identification of retinal diseases.

1.5 Objectives of the Project:

The primary objective of this project is to develop an efficient system for the classification of retinal diseases using Optical Coherence Tomography (OCT) images. The specific objectives of the project are as follows:

- To preprocess OCT images and prepare a structured dataset suitable for training and evaluation.
- To analyze retinal images for distinguishing features associated with different retinal conditions.
- To develop deep learning models capable of accurately classifying retinal diseases.

- To apply transfer learning techniques to improve model performance and reduce training time.
- To design an ensemble approach that enhances classification accuracy and robustness.
- To evaluate the performance of the developed models using appropriate metrics such as accuracy and confusion matrix.
- To visualize model predictions and understand important regions in OCT images.
- To create a system that supports reliable and efficient analysis of retinal images.

1.6 Scope of the Project:

The scope of this project focuses on the classification of retinal diseases using Optical Coherence Tomography (OCT) images. The system is designed to analyze retinal scans and categorize them into four classes: Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), DRUSEN, and NORMAL. It aims to assist in the accurate identification of retinal abnormalities based on structural features present in OCT images.

The project emphasizes the use of deep learning techniques for automated analysis of medical images, reducing the need for manual interpretation. It is intended to support healthcare professionals by providing consistent and efficient analysis of retinal conditions. The system can be used as a decision-support tool in clinical environments where large volumes of imaging data need to be processed.

Furthermore, the scope of the project can be extended to include additional retinal diseases and larger datasets to improve generalization. Future enhancements may involve real-time implementation, integration with clinical systems, and development of user- friendly interfaces for practical healthcare applications. The approach can also be adapted for other types of medical image analysis, making it a scalable solution in the field of artificial intelligence in healthcare.

CHAPTER-2

LITERATURE SURVEY

2.1 Journal 1:

TITLE: Explainable Deep Learning Approach for Automated Classification of Retinal Diseases Using OCT Images and Grad-CAM

AUTHOR: Gurjot Kaur, Raghavendra Sridhar, Rashi Nimesh Kumar Dhenia, Ishva Jitendrakumar Kanani, IEEE (2025)

This paper presents an explainable deep learning framework for retinal disease classification using OCT images. It employs CNN models along with Grad-CAM to provide interpretability of predictions, helping identify important regions influencing decisions. The study demonstrates high accuracy and improved transparency in diagnosis. This work is relevant to the proposed project as it highlights classification and explainability, while the proposed system extends it using ensemble learning for enhanced accuracy and robustness.

2.2 Journal 2:

TITLE: A Lightweight Ensemble Learning with Uncertainty Quantification for Robust OCT Retinal Disease Prediction

AUTHOR: S. Madhumithaa, Masoodhu Banu N.M., et al., IEEE (2025)

This research focuses on the development of a lightweight ensemble learning framework for retinal disease prediction using OCT images. By combining multiple deep learning models, the proposed system improves classification accuracy while maintaining computational efficiency. The study incorporates uncertainty quantification to evaluate the confidence of predictions, enabling the identification of uncertain or unreliable outputs. This approach enhances the reliability and safety of automated medical diagnosis systems. The results demonstrate that ensemble-based methods, along with uncertainty estimation, can significantly improve robustness, generalization, and trust in retinal disease classification tasks.

2.3 Journal 3:

TITLE: Multi-class Retinal OCT Image Analysis Based on EfficientNetV2

AUTHOR: Jipeng Luan, Hongfeng Sun, Shengpeng Yu, Jie Tian, Zhen Cui, Hong Wang, IEEE (2025)

This research focuses on the application of EfficientNetV2 for multi-class retinal disease classification using OCT images. The proposed model utilizes optimized network scaling and advanced feature extraction techniques to accurately identify different retinal conditions. By leveraging the efficiency of EfficientNetV2, the system achieves high classification performance with reduced computational complexity. The study demonstrates that deep learning models can effectively capture fine-grained features from OCT scans, leading to improved accuracy and generalization in medical image analysis tasks.

2.4 Journal 4:

TITLE: Automated Retinal Disease Detection: VGG16-Based Deep Learning Approach for OCT Image Classification

AUTHOR: Pratham Kaushik, Sunila Choudhary, IEEE (2025)

This research focuses on the use of the VGG16 convolutional neural network for automated retinal disease detection using OCT images. The proposed system applies transfer learning to utilize pre-trained weights, enabling efficient feature extraction even with limited datasets. The model captures hierarchical image features through deep convolutional layers, resulting in reliable classification performance. The study demonstrates that VGG16-based approaches provide a simple yet effective solution for medical image classification and can serve as a strong baseline for retinal disease detection systems.

CHAPTER-3

SYSTEM DESIGN

3.1 Overview of Proposed System:

The proposed system is designed to automatically classify retinal diseases using Optical Coherence Tomography (OCT) images through a deep learning-based framework. The system follows a structured pipeline that includes data loading, preprocessing, model training, ensemble prediction, and evaluation. It focuses on classifying retinal conditions into four categories: Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), DRUSEN, and NORMAL.

Initially, OCT images are collected from the dataset and organized into respective classes. These images undergo preprocessing steps such as resizing, normalization, and data augmentation using techniques like flipping, rotation, brightness adjustment, and noise addition. These steps help improve data diversity and enhance the generalization capability of the models.

The processed images are then fed into deep learning models through a custom data generator, which efficiently handles batch loading and on-the-fly augmentation. Multiple pretrained convolutional neural network architectures, including DenseNet121, EfficientNetB4, and ResNet50V2, are utilized for feature extraction and classification. These models are first trained using transfer learning and later fine-tuned to improve performance on OCT data.

To achieve higher accuracy and robustness, the outputs of individual models are combined using a weighted ensemble strategy. This approach leverages the strengths of different models and reduces the limitations of individual architectures. The system performance is evaluated using metrics such as accuracy, confusion matrix, classification report, and ROC curves. Additionally, explainability techniques such as Grad-CAM are used to visualize important regions in the OCT images that influence model predictions. A multi-stage clinical pipeline is also implemented to provide disease classification, confidence estimation, and severity analysis.

Overall, the proposed system provides an efficient and reliable framework for automated retinal disease classification, supporting accurate diagnosis and aiding clinical decision-making.

This Figure 6 below provides an overview of the proposed system for retinal disease classification using OCT images.

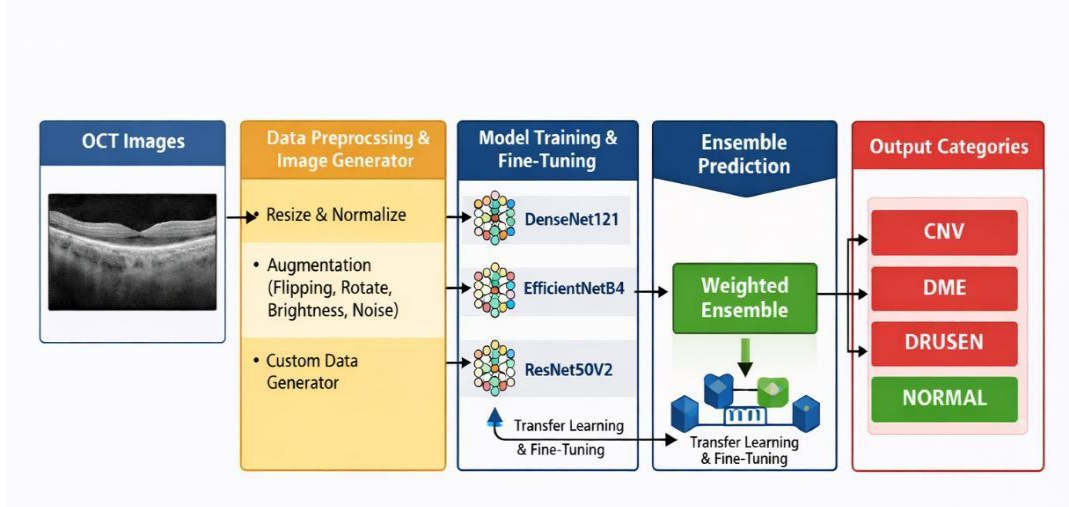


Figure 6: Overview of the Proposed System for Retinal Disease Classification using OCT Images

3.2 Major Components of the System:

The proposed system utilizes multiple deep learning architectures to effectively classify retinal diseases from OCT images. These models are selected based on their strong feature extraction capabilities and proven performance in image classification tasks.

In this system, three pretrained convolutional neural network architectures are used, namely DenseNet121, EfficientNetB4, and ResNet50V2. Each model captures different levels of features from the input images, contributing to improved classification performance.

These models are implemented using transfer learning and fine-tuning techniques, allowing them to adapt to the OCT dataset. The combination of multiple models enhances the robustness and generalization ability of the system.

The following subsections describe each model architecture used in the proposed system.

3.2.1 DenseNet121 Architecture:

DenseNet121 is a convolutional neural network architecture known for its dense connectivity pattern, where each layer is connected to every other layer in a feed-forward manner. This structure allows feature reuse and improves the flow of information and gradients throughout the network. Unlike traditional convolutional networks, DenseNet establishes direct

connections between all layers, ensuring that each layer receives inputs from all preceding layers. This helps in reducing the vanishing gradient problem and enables efficient feature propagation.

The architecture consists of multiple dense blocks and transition layers. Dense blocks contain convolutional layers that are densely connected, while transition layers are used to reduce the spatial dimensions and control model complexity. Fig. 3.2 shows the detailed architecture of DenseNet121, including dense blocks, transition layers, and fully connected layers.

The Figure 7 shows the DenseNet121 architecture with dense blocks and transition layers.

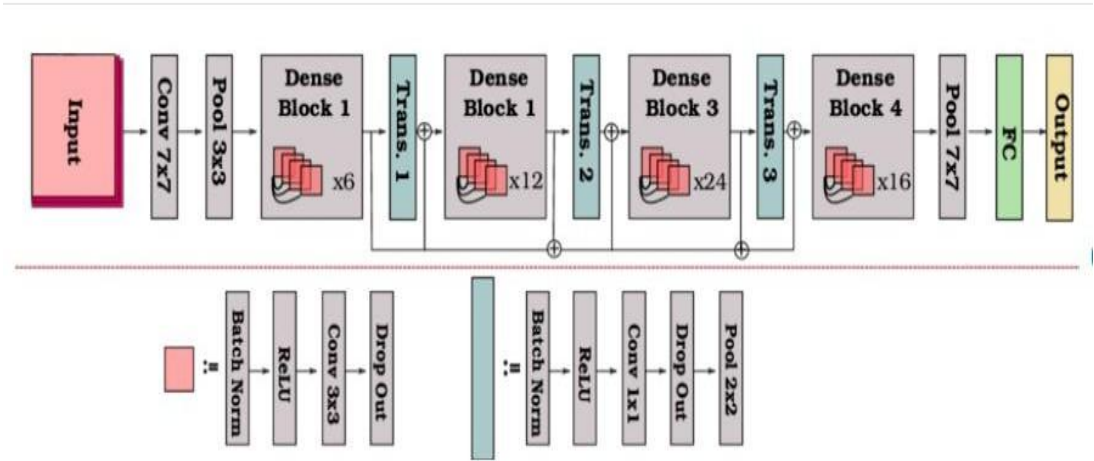


Figure 7: DenseNet121 architecture illustrating dense blocks and transition layers

In this project, DenseNet121 is used as a pretrained model with weights initialized from ImageNet. The final classification layers are modified to suit the retinal disease classification task. Transfer learning is applied by freezing the initial layers and fine-tuning the deeper layers to adapt the model to OCT images. DenseNet121 contributes to the system by effectively capturing fine-grained features in retinal images, which are essential for distinguishing between different disease categories.

3.2.2 EfficientNetB4 Architecture:

EfficientNetB4 is a convolutional neural network architecture that is designed using a compound scaling method to achieve better performance with fewer parameters. Unlike traditional models that scale depth, width, or resolution independently, EfficientNet uniformly scales all three dimensions in a balanced manner. This results in improved accuracy and

efficiency.

The architecture is based on mobile inverted bottleneck convolution (MBConv) blocks combined with squeeze-and-excitation (SE) optimization. These components enable the network to focus on important features while maintaining computational efficiency. The model consists of multiple stages, each containing convolutional layers with varying filter sizes and depths. The following figure 8 represents the EfficientNetB4 architecture with MBConv blocks.

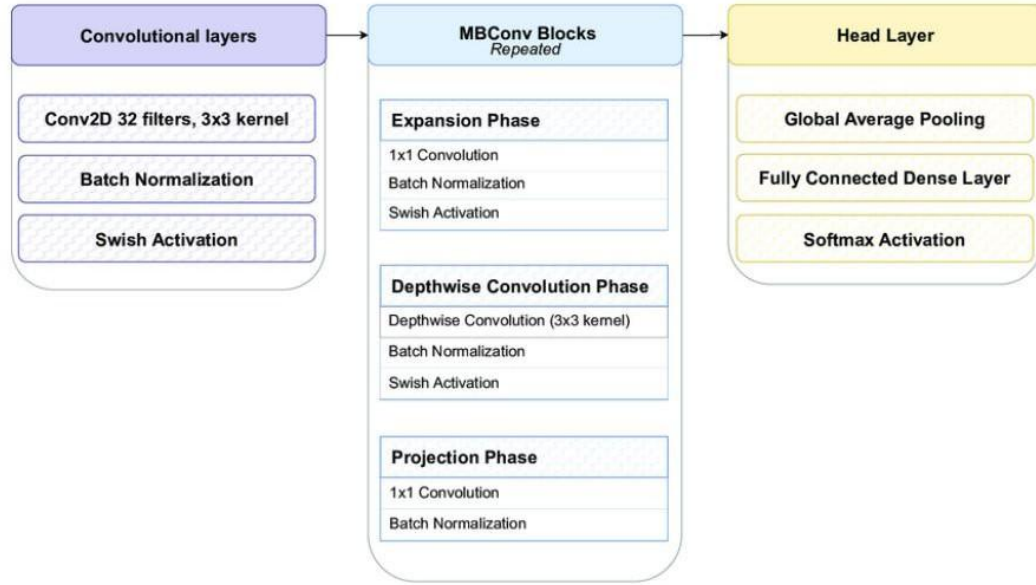


Figure 8: EfficientNetB4 architecture with MBConv blocks

EfficientNetB4 is pretrained on the ImageNet dataset and is adapted for retinal disease classification using transfer learning. In this project, the final layers of the model are modified to classify OCT images into four categories: CNV, DME, DRUSEN, and NORMAL. Fine-tuning is applied to deeper layers to improve performance on the specific dataset.

Due to its efficient scaling and strong feature extraction capability, EfficientNetB4 plays a significant role in capturing both global and local patterns in OCT images, contributing to improved classification accuracy in the proposed system.

3.2.3 ResNet50V2 Architecture:

ResNet50V2 is a deep convolutional neural network architecture that uses residual learning to improve training performance and enable deeper networks. It introduces skip connections, which allow the input of a layer to be directly added to its output. This helps in overcoming the vanishing gradient problem and ensures better gradient flow during training.

The key difference between ResNet50 and ResNet50V2 lies in the arrangement of layers within the residual blocks. In ResNet50V2, batch normalization and activation functions are applied before the convolutional layers, which improves optimization and training stability.

The architecture consists of multiple residual blocks grouped into different stages. Each residual block contains convolutional layers along with shortcut connections that perform identity mapping. These shortcut connections allow the network to learn residual functions instead of direct mappings, improving feature learning efficiency.

In this project, ResNet50V2 is used as a pretrained model with weights initialized from the ImageNet dataset. The final classification layers are modified to suit the OCT retinal disease classification task. Transfer learning is applied by freezing initial layers, followed by fine-tuning of deeper layers to improve performance on the dataset.

ResNet50V2 plays a crucial role in capturing complex structural patterns in OCT images, making it highly effective for accurate classification of retinal diseases.

This Figure 9 shows the ResNet50V2 architecture with residual blocks and skip connections.

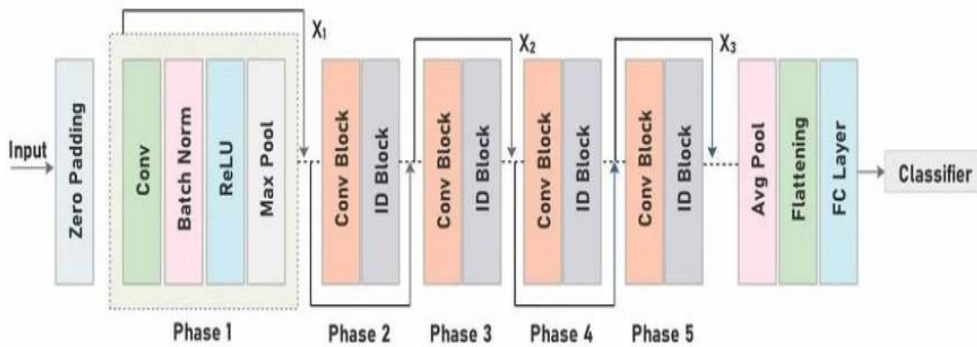


Figure 9: ResNet50V2 architecture with residual blocks connections

3.3 Ensemble Learning Strategy:

The proposed system utilizes an ensemble learning strategy to improve the overall accuracy and robustness of retinal disease classification. Instead of relying on a single model, the system combines the predictions of multiple deep learning architectures, namely DenseNet121, EfficientNetB4, and ResNet50V2. Each model is trained independently on the OCT dataset and learns different feature representations from the input images. Due to variations in architecture and feature extraction capabilities, each model contributes unique strengths to the classification task.

To effectively combine these models, a weighted ensemble approach is used. The weights

are assigned based on the validation performance of each model. Models with higher validation accuracy are given greater importance in the final prediction. The final output is obtained by computing the weighted average of the predicted probabilities from all models.

In addition to ensemble learning, Test-Time Augmentation (TTA) is applied during prediction to further enhance performance. Multiple augmented versions of the same input image, such as flipped and brightness-adjusted images, are passed through the models, and their predictions are averaged. This reduces prediction variance and improves generalization.

The ensemble strategy significantly improves classification performance compared to individual models by reducing errors and increasing prediction reliability. It ensures that the system produces more stable and accurate results, making it suitable for real-world medical image analysis. The final prediction is obtained using a weighted combination of outputs from individual models, which can be represented as:

$$P_{\text{final}} = w_1 P_{\text{DenseNet}} + w_2 P_{\text{EfficientNet}} + w_3 P_{\text{ResNet}}$$

The Figure10 illustrates the ensemble learning strategy combining predictions from multiple models.

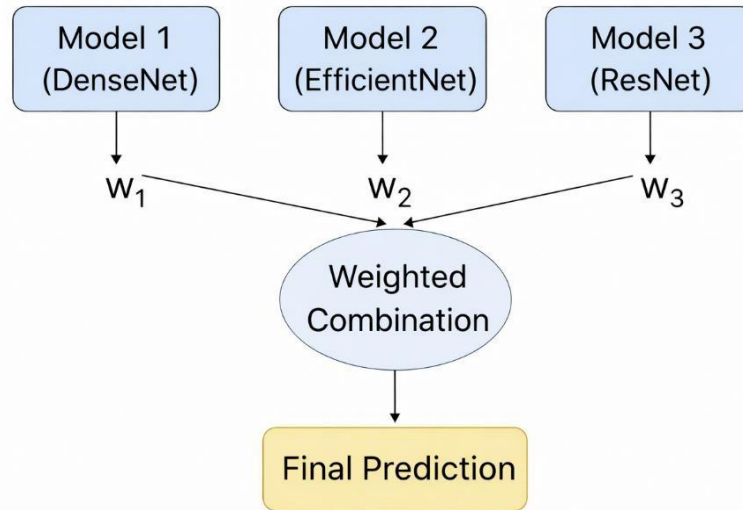


Figure 10: Ensemble learning strategy combining predictions from models

3.4 System Architecture Diagram:

The system architecture represents the overall structure and workflow of the proposed retinal disease classification system. It illustrates how input OCT images are processed through different stages, including preprocessing, model prediction, ensemble learning, and final output generation.

The Figure 11 representation shows the system architecture of the proposed retinal disease classification framework.

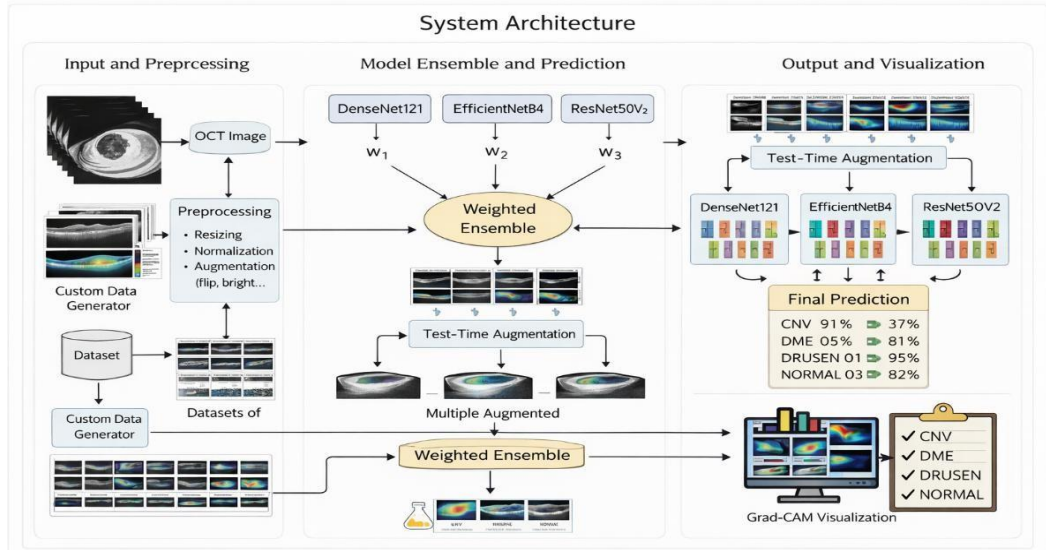


Figure 11: System architecture of the proposed retinal disease classification framework

Initially, OCT images are provided as input to the system. These images undergo preprocessing steps such as resizing, normalization, and data augmentation to improve image quality and ensure consistency. A custom data generator is used to efficiently load and process images in batches during training and testing.

The processed images are then passed through multiple deep learning models, including DenseNet121, EfficientNetB4, and ResNet50V2. These models extract relevant features from the images and generate prediction probabilities for each class.

The outputs from these models are combined using a weighted ensemble strategy, where each model contributes to the final prediction based on its performance. Additionally, Test-Time Augmentation (TTA) is applied during inference to improve prediction stability by averaging results from augmented versions of the same image.

The final output of the system includes the predicted class (CNV, DME, DRUSEN, or NORMAL), along with confidence scores. The system also incorporates visualization techniques such as Grad-CAM to highlight important regions in the OCT images, aiding interpretability.

CHAPTER-4

SOFTWARE DESIGN

4.1 Software Requirements:

The software requirements define the tools, programming environments, and libraries necessary for the development and execution of the Smart Traffic Management System for Urban Congestion. Since the project is entirely software-based, the selection of efficient and compatible software resources is crucial to ensure smooth functioning, high performance, and scalability.

4.1.1 Operating System:

- Windows 10 / 11 or Linux (Ubuntu 20.04 or higher):

The proposed system is compatible with both Windows and Linux operating systems. Windows provides a user-friendly development environment with tools such as VS Code and Jupyter Notebook, making it suitable for initial development and testing.

Linux-based systems, particularly Ubuntu, offer better support and compatibility for deep learning frameworks like TensorFlow and efficient handling of large-scale datasets.

In this project, a cloud-based Linux environment (Kaggle) is primarily used, which provides GPU acceleration and a stable platform for training deep learning models on OCT retinal image data.

4.1.2 Programming Language:

- Python 3.10 or above:

Python is used as the primary programming language for developing the proposed system due to its simplicity, flexibility, and extensive support for machine learning and deep learning applications. It provides a rich ecosystem of libraries and frameworks such as TensorFlow, Keras, NumPy, and OpenCV, which are essential for building and training convolutional neural networks (CNNs).

Python also enables efficient handling of large-scale medical image datasets, implementation of preprocessing techniques, and integration of advanced methods such as transfer learning, fine-tuning, and ensemble learning.

Its compatibility with platforms like Kaggle further facilitates seamless model training using GPU acceleration and simplifies the overall development process.

4.1.3 Libraries:

The proposed system utilizes several Python libraries to implement data preprocessing, model training, evaluation, and visualization tasks efficiently. The key libraries used in this project are as follows:

- **TensorFlow & Keras:**

TensorFlow is the primary deep learning framework used for building, training, and evaluating convolutional neural network (CNN) models. Keras, integrated within TensorFlow, provides a high-level API for implementing pre-trained architectures such as DenseNet121, EfficientNetB4, and ResNet50V2. It also supports transfer learning, fine-tuning, and model optimization.

- **NumPy:**

NumPy is used for numerical computations and efficient handling of multi-dimensional arrays. It plays a crucial role in processing image data, managing tensors, and performing mathematical operations during model training and evaluation.

- **OpenCV (cv2):**

OpenCV is used for image processing operations such as resizing, normalization, and enhancement of OCT images. It helps in preparing the input data before feeding it into deep learning models.

- **Matplotlib:**

Matplotlib is used for visualizing results such as training and validation accuracy, loss curves, confusion matrix, and performance comparisons between different models.

- **Pandas:**

Pandas is used for handling structured data and organizing results such as classification reports, accuracy comparisons, and dataset summaries.

- **Scikit-learn:**

Scikit-learn is used for performance evaluation metrics including accuracy, precision, recall, F1-score, and confusion matrix generation, which are essential for assessing model performance.

- **OS & Glob:**

These libraries are used for file handling, directory management, and dataset path navigation, enabling efficient loading and organization of OCT image data.

4.1.4 Development Requirements(Kaggle):

- **Kaggle:**

Kaggle is used as the primary development environment for implementing the proposed system. It provides a cloud-based platform with integrated support for Python, Jupyter Notebooks, and pre-installed machine learning libraries such as TensorFlow and OpenCV. Kaggle offers free access to GPU acceleration, which significantly improves the training speed of deep learning models. It also allows efficient handling of large- scale datasets, making it suitable for processing OCT retinal images.

Additionally, Kaggle ensures a reproducible and well-organized workflow by enabling easy dataset management, version control, and seamless execution of the complete deep learning pipeline, including preprocessing, model training, evaluation, and visualization.

4.2 Software Design Overview:

The software design of the proposed system is developed as a structured pipeline for automated retinal disease classification using OCT images. The system begins with data preprocessing, where input images are resized, normalized, and augmented using techniques such as flipping, rotation, brightness adjustment, and noise addition. A custom data generator is used to efficiently load and process images in batches during training. The preprocessed data is then fed into deep learning models, including DenseNet121, EfficientNetB4, and ResNet50V2, which utilize transfer learning and fine-tuning to extract meaningful features and perform classification.

The predictions from individual models are combined using a weighted ensemble strategy to improve overall accuracy and robustness. Additionally, Test-Time Augmentation (TTA) is applied to enhance prediction stability. The system also incorporates explainability through Grad-CAM, which highlights important regions in OCT images influencing model decisions. Finally, the system outputs the predicted class along with confidence scores, providing an efficient and reliable framework for retinal disease analysis. The Figure 12 outlines the software design workflow of the proposed deep learning system.

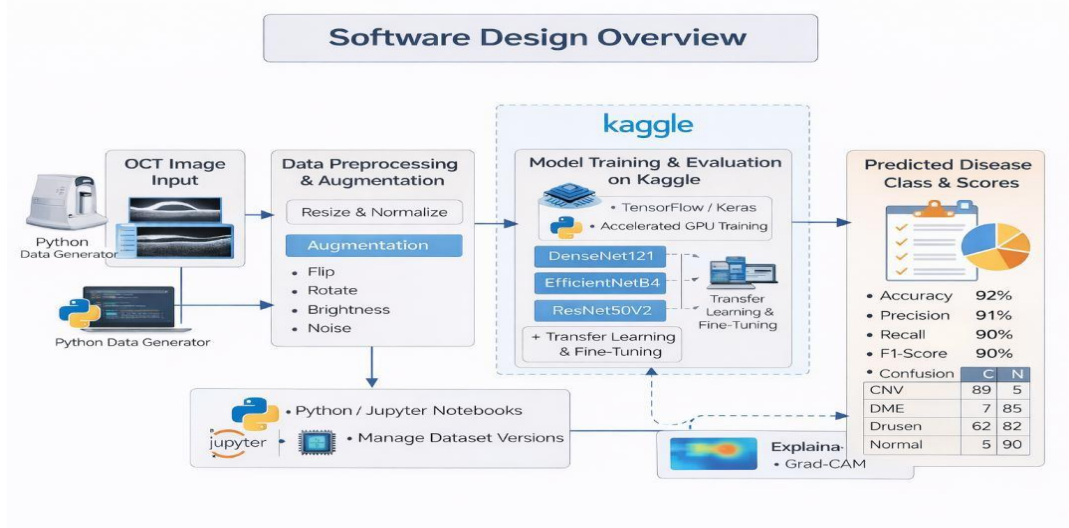


Figure 12: Software Design Workflow of the Proposed Deep Learning System

4.3 Module Description:

4.3.1 Data Preprocessing & Augmentation:

Data preprocessing is performed to standardize OCT images before feeding them into deep learning models. All input images are resized to a fixed resolution and normalized to ensure uniform pixel value distribution. A custom data generator is implemented using TensorFlow/Keras to efficiently load images in batches during training.

To improve generalization and prevent overfitting, data augmentation techniques are applied. These include random horizontal flipping, rotation, brightness adjustment, and noise addition. This process increases dataset diversity and helps the models learn robust features from retinal images.

4.3.2 Model Training & Fine-tuning:

The system utilizes multiple deep learning architectures, including DenseNet121, EfficientNetB4, and ResNet50V2, for retinal disease classification. These models are initialized with pre-trained weights (ImageNet) and adapted using transfer learning.

Initially, base layers are frozen to retain learned features, and custom classification layers are added. Subsequently, fine-tuning is performed by unfreezing selected top layers to improve performance on OCT data. The models are trained using optimized hyperparameters, ensuring efficient convergence and improved accuracy.

4.3.3 Ensemble Prediction Module:

To enhance prediction performance, an ensemble learning strategy is implemented by combining outputs from multiple trained models. Each model generates probability scores for different classes, and these predictions are aggregated using weighted averaging.

Additionally, Test-Time Augmentation (TTA) is applied to improve prediction stability by averaging outputs from augmented versions of the same input image. The ensemble approach reduces individual model bias and improves overall classification accuracy.

4.3.4 Ensemble Prediction Module:

To improve model transparency, Grad-CAM (Gradient-weighted Class Activation Mapping) is used for visual explanation of predictions. This technique highlights important regions in OCT images that contribute to the model's decision.

Grad-CAM generates heatmaps that overlay on input images, enabling clinicians to interpret model behavior. This improves trust in the system and provides valuable insights into disease-related features within retinal images.

CHAPTER-5

METHODOLOGY

5.1 Dataset Description (OCT C8 Subset):

The dataset used in this study is derived from a publicly available OCT retinal image dataset, focusing on four major classes: Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), Drusen, and Normal. These categories represent common retinal conditions identifiable through Optical Coherence Tomography (OCT) imaging and are widely used in automated disease diagnosis research. The images are grayscale OCT scans that capture detailed retinal layer structures, enabling effective detection of pathological abnormalities.

The dataset is divided into training, validation, and testing subsets to ensure reliable model development and evaluation. Each class is uniformly distributed across all subsets, preventing class imbalance and improving the generalization capability of the model. The training set is used for learning, the validation set for monitoring and tuning during training, and the test set for final performance evaluation, ensuring unbiased and accurate results.

This Figure13 shows the class-wise distribution of the OCT dataset across training, validation, and testing sets.

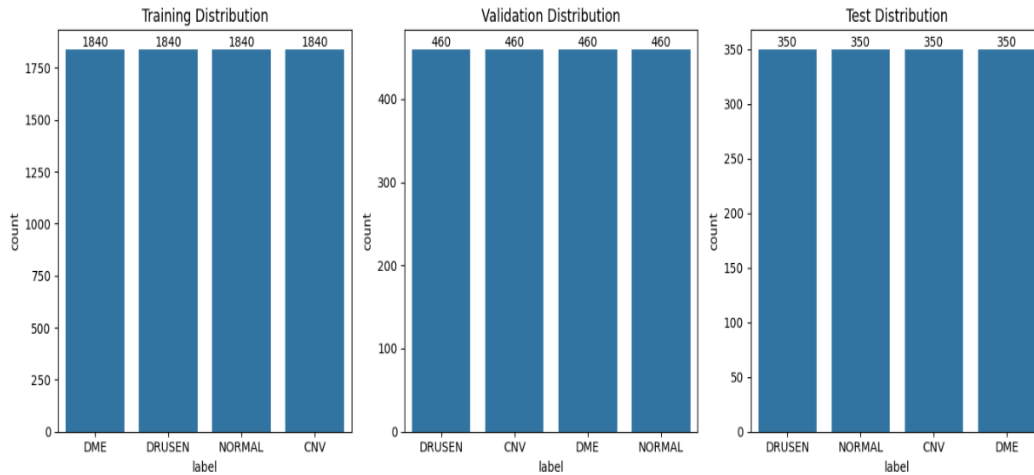


Figure 13: Class-wise distribution of OCT dataset across three sets

5.2 Data Preprocessing Pipeline:

The data preprocessing pipeline is designed to enhance the quality and consistency of OCT retinal images before feeding them into deep learning models. Initially, all images are resized to a fixed input size to match the requirements of pre-trained CNN architectures. The

images are then normalized by scaling pixel intensity values, which helps in improving convergence and stabilizing the training process.

To increase model robustness and prevent overfitting, various data augmentation techniques are applied, including horizontal flipping, rotation, brightness adjustment, and noise addition. A custom data generator is implemented to efficiently load and preprocess images in batches during training. This ensures that the model is exposed to diverse variations of OCT images, thereby improving its generalization performance on unseen data.

The Figure 14 depicts the data preprocessing pipeline including original, preprocessed, and augmented OCT images.

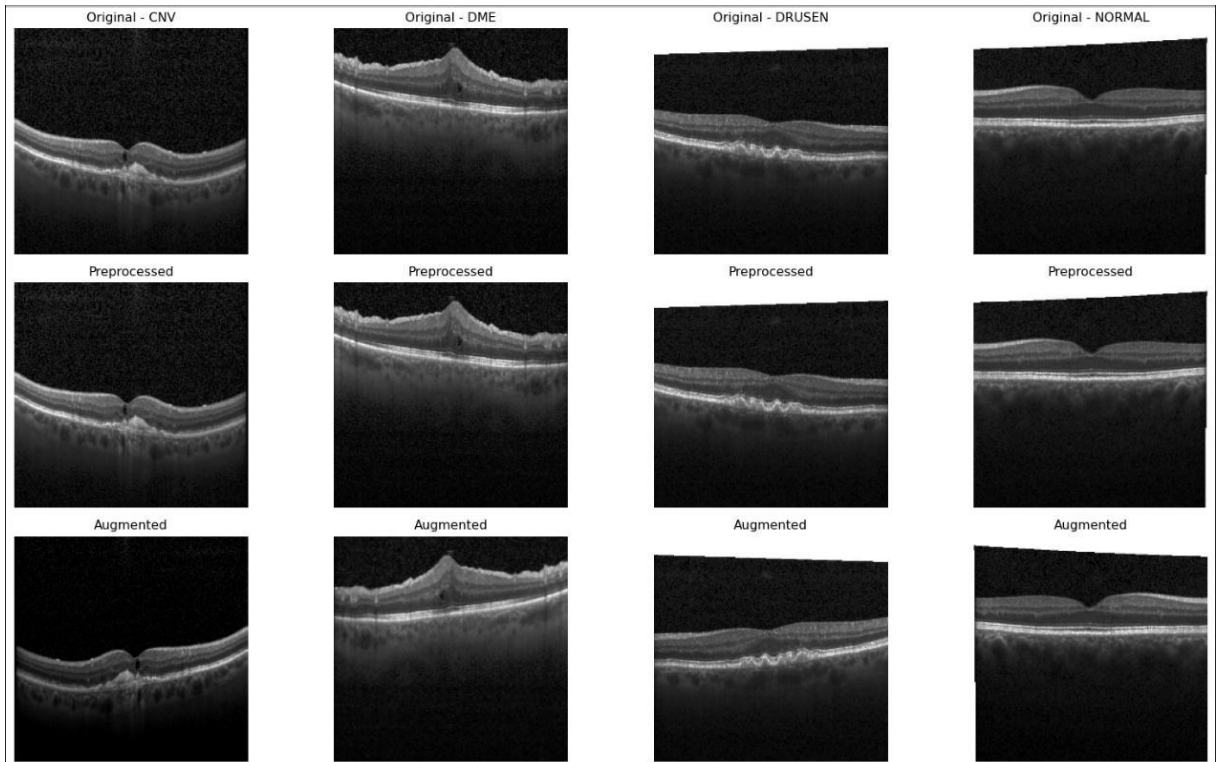


Figure 14: Data preprocessing pipeline showing original, preprocessed, and augmented OCT images

5.3 Model Training Strategy:

The model training strategy employs transfer learning using pre-trained convolutional neural networks, namely DenseNet121, EfficientNetB4, and ResNet50V2, to extract meaningful features from OCT retinal images. These models are initialized with ImageNet weights, and their final layers are modified to suit the four-class classification task. Fine-tuning is performed by unfreezing the top layers to adapt the models to domain-specific features. The models are trained using categorical cross-entropy loss and optimized with the Adam optimizer,

along with techniques such as early stopping and learning rate scheduling to prevent overfitting and ensure efficient convergence.

The Figure 15 shows the transfer learning and fine-tuning process for model training.

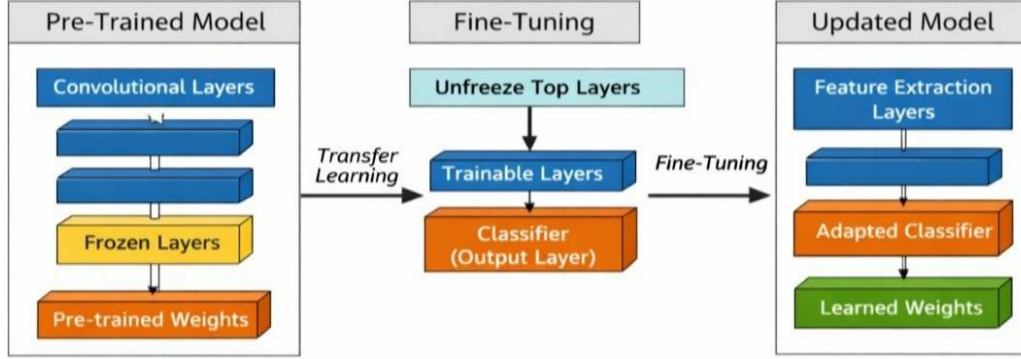


Figure 15: Transfer Learning and Fine-Tuning Process for Model Training

5.4 Ensemble Learning Approach:

The ensemble learning approach combines the predictions of multiple deep learning models, namely DenseNet121, EfficientNetB4, and ResNet50V2, to improve overall classification performance. Each model generates probability scores for the input OCT image, and these outputs are combined using a weighted averaging technique based on model performance. Additionally, Test-Time Augmentation (TTA) is applied by evaluating multiple augmented versions of the same image and averaging their predictions. This approach reduces individual model bias, enhances robustness, and improves the overall accuracy and reliability of the system.

The Figure 16 presents the test-time augmentation (TTA) and ensemble learning approach.

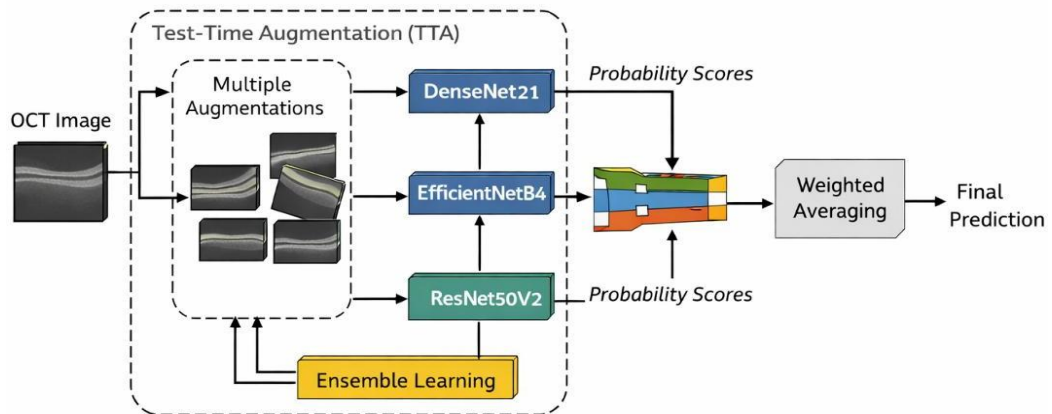


Figure 16: TTA and Ensemble Learning Approach

5.5 Evaluation Metrics:

The performance of the proposed system is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of the model's ability to correctly classify retinal diseases. Additionally, a confusion matrix is used to visualize class-wise performance and identify misclassifications. These evaluation measures ensure a reliable and detailed analysis of the model's effectiveness on the OCT dataset.

The Figure 17 illustrates the confusion matrix representing classification performance.

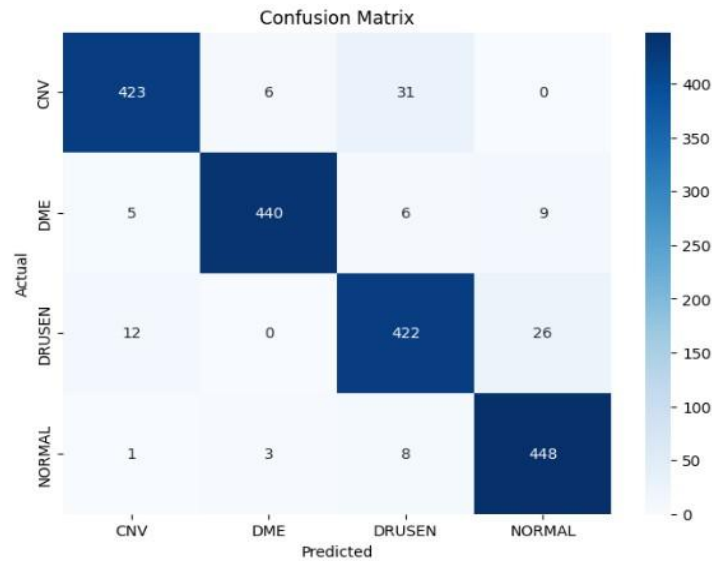


Figure 17: Confusion matrix illustrating classification performance

5.6 Explainability using Grad-CAM:

To enhance interpretability, Grad-CAM (Gradient-weighted Class Activation Mapping) is used to visualize the regions of OCT images that influence the model's predictions. This technique generates heatmaps highlighting important features learned by the model, enabling better understanding of decision-making. Grad-CAM improves transparency and helps clinicians trust the system by providing visual evidence for the predicted class.

The Figure 18 highlights Grad-CAM visualization showing important regions influencing model prediction.

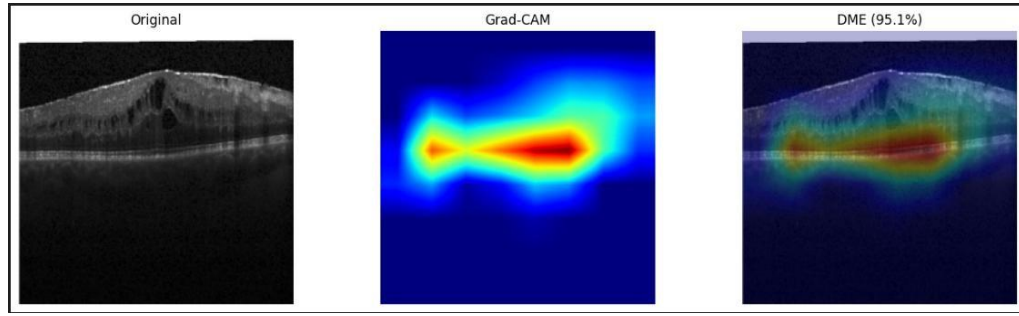


Figure 18: Grad-CAM visualization highlighting important region Influencing model prediction

5.1 Multi-Stage Clinical Decision Pipeline:

The proposed system follows a multi-stage clinical decision pipeline that integrates preprocessing, model prediction, ensemble learning, and explainability. OCT images are first preprocessed and passed through multiple trained models, whose predictions are combined using an ensemble approach. The final output includes the predicted disease class along with confidence scores, supported by Grad-CAM visualizations. This structured pipeline ensures accurate, reliable, and clinically interpretable results for retinal disease diagnosis.

5.2 Block Diagram:

The block diagram shows the workflow of the OCT-based retinal disease detection system. It includes preprocessing, model prediction using multiple deep learning models, and ensemble learning for improved accuracy. The system finally performs disease classification, severity estimation, and evaluation. The Figure 19 represents the block diagram of OCT retinal disease detection.

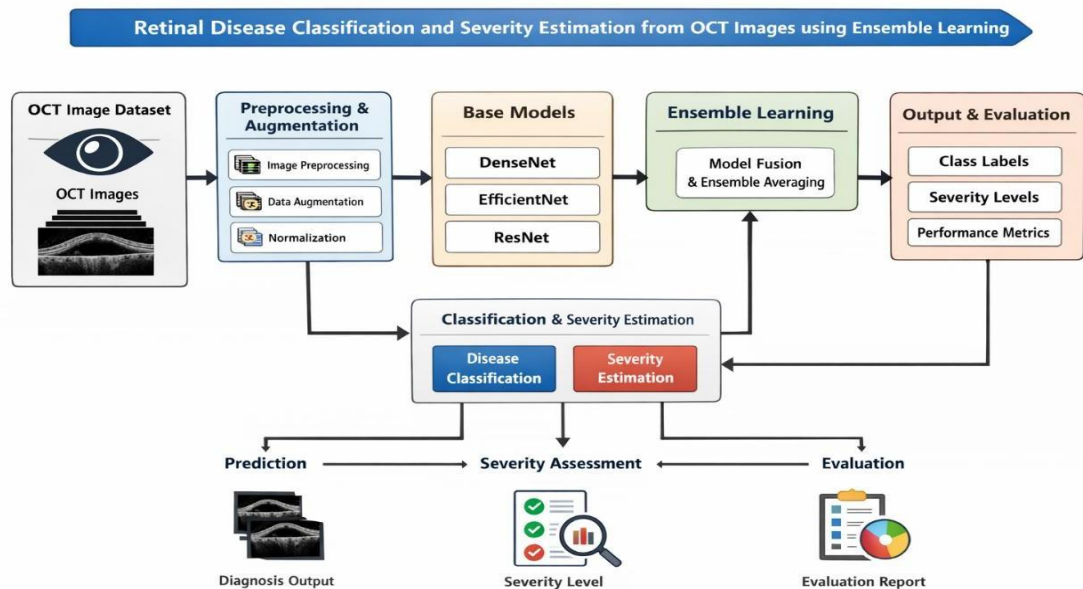


Figure 19: Block Diagram of OCT retinal disease detection

CHAPTER-6

SOURCE CODE

6.1 Data Preprocessing.py:

```
import os
import glob
import random
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import cv2
import warnings
warnings.filterwarnings('ignore')
import tensorflow as tf
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
import albumentations as A
class Config:
    BASE_PATH="/kaggle/input/datasets/obulisainaren/retinal-oct
c8/RetinalOCT_Dataset/RetinalOCT_Dataset"
    IMG_SIZE = (256, 256)
    BATCH_SIZE = 32
    CLASSES = ['CNV', 'DME', 'DRUSEN', 'NORMAL']
cfg = Config()
random.seed(42)
np.random.seed(42)
tf.random.set_seed(42)
def load_data(base_path, split):
    paths, labels = [], []
    split_path = os.path.join(base_path, split)
```

```

for cls in cfg.CLASSES:
    cls_folder = os.path.join(split_path, cls)
    images = glob.glob(os.path.join(cls_folder, "*.jpg")) + \
        glob.glob(os.path.join(cls_folder, "*.png")) + \
        glob.glob(os.path.join(cls_folder, "*.jpeg"))
    paths.extend(images)
    labels.extend([cls]*len(images))
return paths, labels
train_paths, train_labels = load_data(cfg.BASE_PATH, "train")
test_paths, test_labels = load_data(cfg.BASE_PATH, "test")
df = pd.DataFrame({'path': train_paths, 'label': train_labels})
train_paths, val_paths, train_labels, val_labels = train_test_split(
    df['path'].values,
    df['label'].values,
    test_size=0.2,
    stratify=df['label'].values,
    random_state=42
)
df_train = pd.DataFrame({'path': train_paths, 'label': train_labels})
df_val = pd.DataFrame({'path': val_paths, 'label': val_labels})
df_test = pd.DataFrame({'path': test_paths, 'label': test_labels})
transform = A.Compose([
    A.HorizontalFlip(p=0.5),
    A.VerticalFlip(p=0.3),
    A.Rotate(limit=15, p=0.5),
    A.RandomBrightnessContrast(p=0.5),
    A.GaussianBlur(p=0.2),
    A.CLAHE(p=0.3)
])
def process_image(path, augment=False):
    img = cv2.imread(path)
    if img is None:

```

```

        return np.zeros((*cfg.IMG_SIZE,3), dtype=np.float32)
    img = cv2.resize(img, cfg.IMG_SIZE)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    if augment:
        img = transform(image=img)["image"]

    img = img.astype(np.float32) / 255.0
    return img

le = LabelEncoder()
le.fit(cfg.CLASSES)
class OCTGenerator(tf.keras.utils.Sequence):
    def __init__(self, df, batch_size=32, shuffle=True, augment=False):
        self.df = df.reset_index(drop=True)
        self.batch_size = batch_size
        self.shuffle = shuffle
        self.augment = augment
        self.indices = np.arange(len(self.df))
        self.labels_encoded = le.transform(self.df["label"].values)
        self.on_epoch_end()
    def __len__(self):
        return int(np.ceil(len(self.df) / self.batch_size))
    def on_epoch_end(self):
        if self.shuffle:
            np.random.shuffle(self.indices)
    def __getitem__(self, idx):
        inds = self.indices[idx*self.batch_size:(idx+1)*self.batch_size]
        X, y = [], []
        for i in inds:
            row = self.df.iloc[i]
            img = process_image(row["path"], augment=self.augment)
            X.append(img)
            y.append(self.labels_encoded[i])

```

```

X = np.array(X, dtype=np.float32)
y = tf.keras.utils.to_categorical(y, num_classes=len(cfg.CLASSES))
return X, y
train_gen = OCTGenerator(df_train,
batch_size=cfg.BATCH_SIZE, augment=True)
val_gen = OCTGenerator(df_val, batch_size=cfg.BATCH_SIZE, augment=False)
test_gen = OCTGenerator(df_test,
batch_size=cfg.BATCH_SIZE, augment=False)
X_batch, y_batch = train_gen[0]
fig, axes = plt.subplots(2, 4, figsize=(12,6))
for i in range(4):
    axes[0, i].imshow(X_batch[i])
    axes[0, i].set_title(cfg.CLASSES[np.argmax(y_batch[i])])
    axes[0, i].axis("off")
    axes[1, i].imshow(X_batch[i+4])
    axes[1, i].set_title(cfg.CLASSES[np.argmax(y_batch[i+4])])
    axes[1, i].axis("off")
plt.suptitle("Generator Samples")
plt.tight_layout()
plt.show()

```

6.2 Model_Training.py:

```

import os
os.environ["TF_CPP_MIN_LOG_LEVEL"] = "3"
import numpy as np
import tensorflow as tf
from tensorflow.keras.applications import DenseNet121, EfficientNetB4,
ResNet50V2
from tensorflow.keras.applications.densenet import preprocess_input as den_pre
from tensorflow.keras.applications.efficientnet import preprocess_input as eff_pre
from tensorflow.keras.applications.resnet_v2 import preprocess_input as res_pre
from tensorflow.keras.models import Model

```

```

from tensorflow.keras.layers import Dense, GlobalAveragePooling2D, Dropout,
BatchNormalization, Multiply
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau,
ModelCheckpoint
from tensorflow.keras.regularizers import l2
from sklearn.utils.class_weight import compute_class_weight
INPUT_SHAPE = (256,256,3)
NUM_CLASSES = 4
class PreprocessWrapper(tf.keras.utils.Sequence):
    def __init__(self, generator, preprocess_func):
        self.generator = generator
        self.preprocess_func = preprocess_func
    def __len__(self):
        return len(self.generator)
    def __getitem__(self, idx):
        X, y = self.generator[idx]
        X = self.preprocess_func((X * 255.0).astype(np.float32))
        return X, y
def se_block(x):
    ch = int(x.shape[-1])
    se = Dense(ch // 8, activation="relu")(x)
    se = Dense(ch, activation="sigmoid")(se)
    return Multiply()([x, se])
def build_model(base_fn):
    base=base_fn(weights="imagenet",include_top=False,input_shape=INPUT_SHA
PE)
    base.trainable = False
    x = base.output
    x = GlobalAveragePooling2D()(x)
    x = BatchNormalization()(x)
    x = Dense(1024, activation="relu", kernel_regularizer=l2(1e-4))(x)

```

```

x = Dropout(0.5)(x)
x = Dense(512, activation="relu")(x)
x = Dropout(0.3)(x)
x = se_block(x)
out = Dense(NUM_CLASSES, activation="softmax")(x)
model = Model(base.input, out)
model.compile(
    optimizer=Adam(1e-4),
    loss=tf.keras.losses.CategoricalCrossentropy(label_smoothing=0.1),
    metrics=["accuracy"])
return model, base

labels = []
for i in range(len(train_gen)):
    _, y = train_gen[i]
    labels.extend(np.argmax(y, axis=1))
labels = np.array(labels)
class_weights = compute_class_weight(
    class_weight="balanced",
    classes=np.unique(labels),
    y=labels)
class_weights = dict(enumerate(class_weights))
def train_model(name, base_fn, pre_fn, unfreeze_n):
    model, base = build_model(base_fn)
    train_w = PreprocessWrapper(train_gen, pre_fn)
    val_w = PreprocessWrapper(val_gen, pre_fn)
    callbacks = [
        EarlyStopping(monitor="val_loss", patience=5, restore_best_weights=True),
        ReduceLROnPlateau(monitor="val_loss", factor=0.3, patience=2),
        ModelCheckpoint(f"/kaggle/working/{name}.keras", save_best_only=True)]
    model.fit(
        train_w,
        validation_data=val_w,

```



```

        epochs=8,
        class_weight=class_weights,
        callbacks=callbacks,
        verbose=1 )
    base.trainable = True
    for layer in base.layers[::-unfreeze_n]:
        layer.trainable = False
    model.compile(
        optimizer=Adam(1e-5),
        loss=tf.keras.losses.CategoricalCrossentropy(label_smoothing=0.1),
        metrics=["accuracy"])
    model.fit(
        train_w,
        validation_data=val_w,
        epochs=25,
        initial_epoch=8,
        class_weight=class_weights,
        callbacks=callbacks,
        verbose=1)
    return model
models_cfg = {
    "DenseNet": (DenseNet121, den_pre, 120),
    "EffNet": (EfficientNetB4, eff_pre, 150),
    "ResNet": (ResNet50V2, res_pre, 100) }
trained = {}
for name, (fn, pre, unf) in models_cfg.items():
    trained[name] = train_model(name, fn, pre, unf)

```

6.3 Ensemble_Model.py:

```

import numpy as np

import tensorflow as tf

import cv2

```

```

from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

CLASS_NAMES=["CNV","DME","DRUSEN","NORMAL"]

def tta_predict_batch(model,X,pre):

    preds=[]

    X1=pre((X*255.0).astype(np.float32))

    preds.append(model.predict(X1,verbose=0))

    X_flip=np.flip(X,axis=2)

    X2=pre((X_flip*255.0).astype(np.float32))

    preds.append(model.predict(X2,verbose=0))

    X_bright=np.clip(X*1.1,0,1)

    X3=pre((X_bright*255.0).astype(np.float32))

    preds.append(model.predict(X3,verbose=0))

    return np.mean(preds,axis=0)

def get_preds(model,gen,pre):

    preds=[]

    for i in range(len(gen)):

        X,_=gen[i]

        p=tta_predict_batch(model,X,pre)

        preds.append(p)

    return np.vstack(preds)

p_dense=get_preds(trained["DenseNet"],test_gen,den_pre)

p_eff=get_preds(trained["EffNet"],test_gen,eff_pre)

p_res=get_preds(trained["ResNet"],test_gen,res_pre)

weights=np.array([0.3,0.3,0.4])

final=weights[0]*p_dense+weights[1]*p_eff+weights[2]*p_res

y_true=[]

```

```

for i in range(len(test_gen)):
    _,y=test_gen[i]
    y_true.extend(np.argmax(y,axis=1))
y_true=np.array(y_true)
    y_pred=np.argmax(final,axis=1)
    dense_acc=accuracy_score(y_true,np.argmax(p_dense,axis=1))
    eff_acc=accuracy_score(y_true,np.argmax(p_eff,axis=1))
    res_acc=accuracy_score(y_true,np.argmax(p_res,axis=1))
    ens_acc=accuracy_score(y_true,y_pred)
    print("DenseNet:",dense_acc)
    print("EffNet:",eff_acc)
    print("ResNet:",res_acc)
    print("Ensemble:",ens_acc)
    print(classification_report(y_true,y_pred))
    cm=confusion_matrix(y_true,y_pred)
    print(cm)
    model=trained["ResNet"]
    def find_last_conv_layer(model):
        for layer in reversed(model.layers):
            try:
                if len(layer.output.shape)==4:
                    return layer.name
            except:
                continue
    last_conv=find_last_conv_layer(model)
    defgradcam(img):
        grad_model=tf.keras.models.Model([model.input],[model.get_layer(last_conv).ou
tput,model.output])

```

```

img=res_pre((img*255.0).astype(np.float32))
with tf.GradientTape() as tape:
    conv_out,preds=grad_model(img)
    idx=tf.argmax(preds[0])
    loss=preds[:,idx]
grads=tape.gradient(loss,conv_out)
pooled=tf.reduce_mean(grads,axis=(0,1,2))
conv_out=conv_out[0]
heatmap=tf.reduce_sum(conv_out*pooled,axis=-1)
heatmap=tf.maximum(heatmap,0)
heatmap/=tf.reduce_max(heatmap)+1e-8
return heatmap.numpy(),int(idx.numpy()),preds.numpy()[0]

def generate_gradcam_sample(path):
    img=cv2.imread(path,cv2.IMREAD_GRAYSCALE)
    img=cv2.resize(img,(224,224))
    img=img.astype(np.float32)/255.0
    img=np.stack([img]*3,axis=-1)
    inp=np.expand_dims(img,axis=0)
    heatmap,pred_idx,probs=gradcam(inp)
    heatmap=cv2.resize(heatmap,(224,224))
    heatmap=np.uint8(255*heatmap)
    heatmap_color=cv2.applyColorMap(heatmap,cv2.COLORMAP_JET)
    heatmap_color=cv2.cvtColor(heatmap_color,cv2.COLOR_BGR2RGB)
    overlay=cv2.addWeighted((img*255).astype(np.uint8),0.6,heatmap_color,0.4,0)
    return overlay,CLASS_NAMES[pred_idx],np.max(probs)

```

CHAPTER-7

RESULTS

7.1 Performance Evaluation:

This bar chart the final accuracy of individual models and the ensemble model on OCT images. DenseNet121, EfficientNetB4, and ResNet50V2 perform well, but the ensemble model achieves the highest accuracy. The Figure 20 shows that combining models improves overall classification performance.

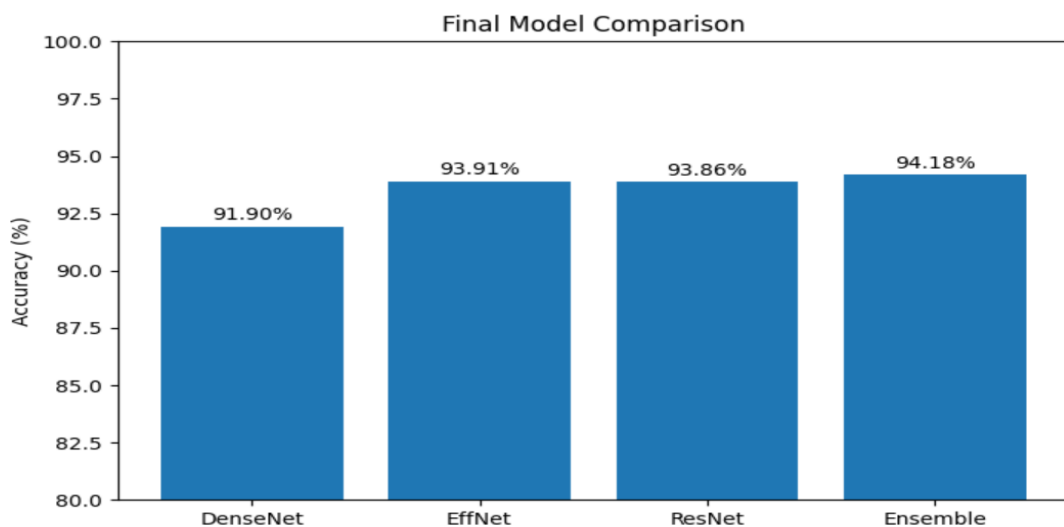


Figure 20: Final Model Comparison

7.2 Model Comparison Results:

This bar chart compares the performance of individual models and highlights the improvement achieved using ensemble learning. DenseNet captures fine features, EfficientNet extracts balanced features, and ResNet learns deep structural patterns. Combining these features, the ensemble model provides more accurate and reliable predictions.

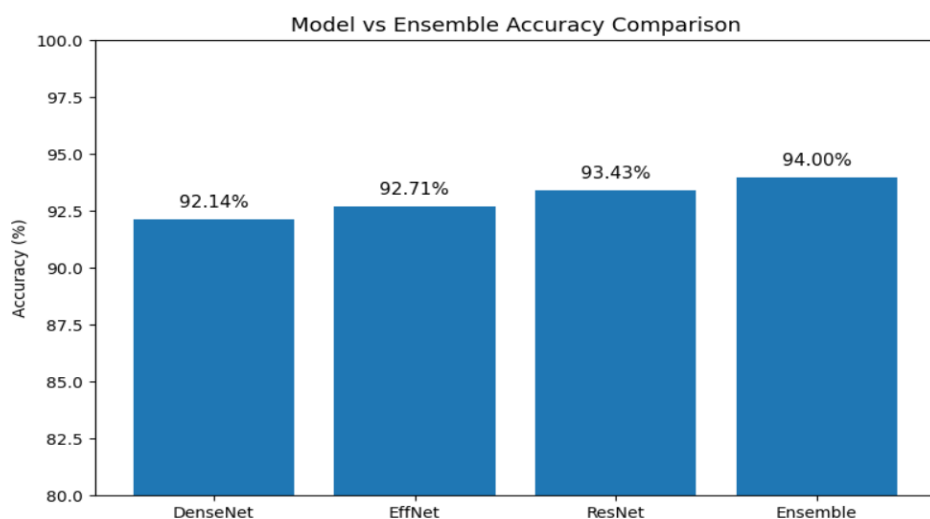


Figure 21: Model Performance Comparison

7.3 Confusion Matrix Analysis:

The confusion matrix analysis shows that the model effectively classifies retinal diseases, with most values concentrated along the diagonal indicating correct predictions and very few misclassifications. It also demonstrates strong performance on unseen test data, confirming good generalization.

7.3.1 Validation Confusion Matrix:

The below Figure 22 Validation confusion matrix shows validation performance with high diagonal values indicating accurate classification.

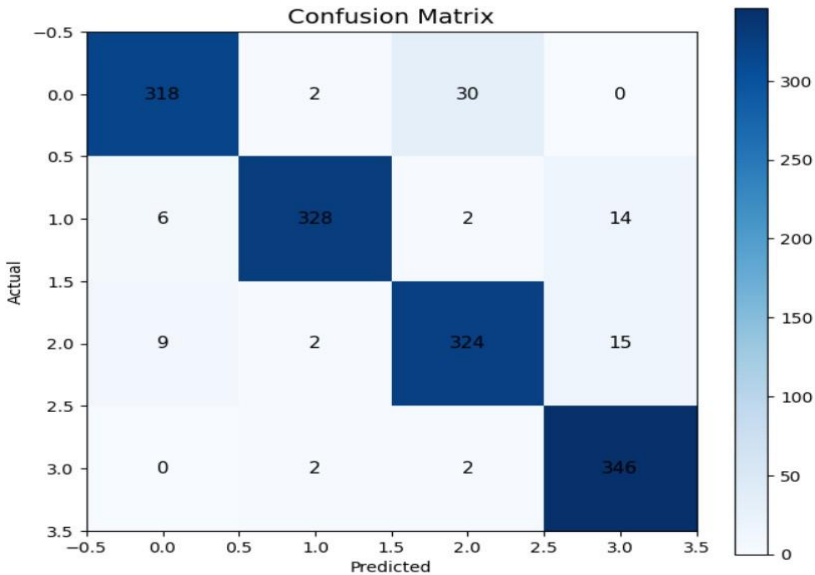


Figure 22: Validation Confusion Matrix

7.3.2 Testing Confusion Matrix:

The below Figure 23 Testing confusion matrix shows testing performance with high diagonal values indicating strong generalization.

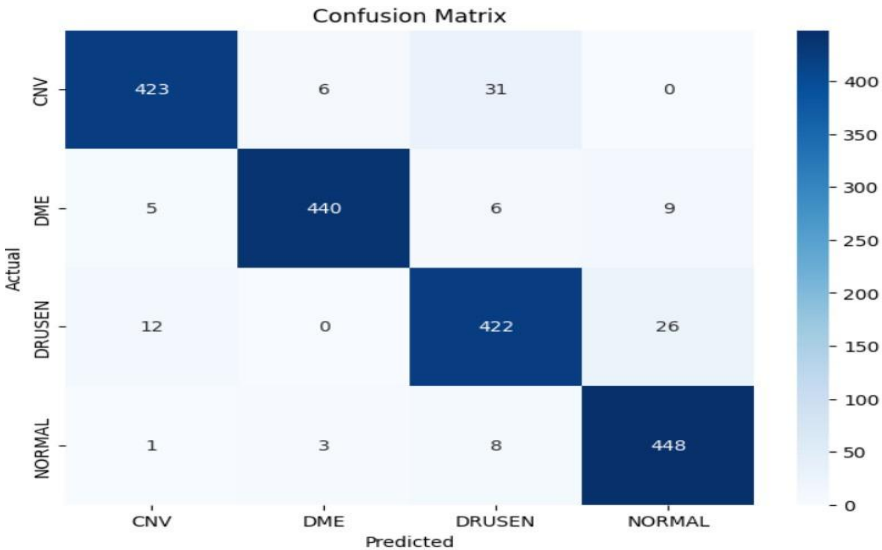


Figure 23: Testing Confusion Matrix

7.4 ROC Curve and Accuracy Analysis:

The ROC and Precision-Recall curves show strong model performance with high true positive rates, precision, and recall. The ensemble model achieves the highest AUC, indicating better discrimination and stability than individual models. Overall, the results confirm improved accuracy and reliability using ensemble learning.

The below Figure 24 shows ROC Curve model comparison.

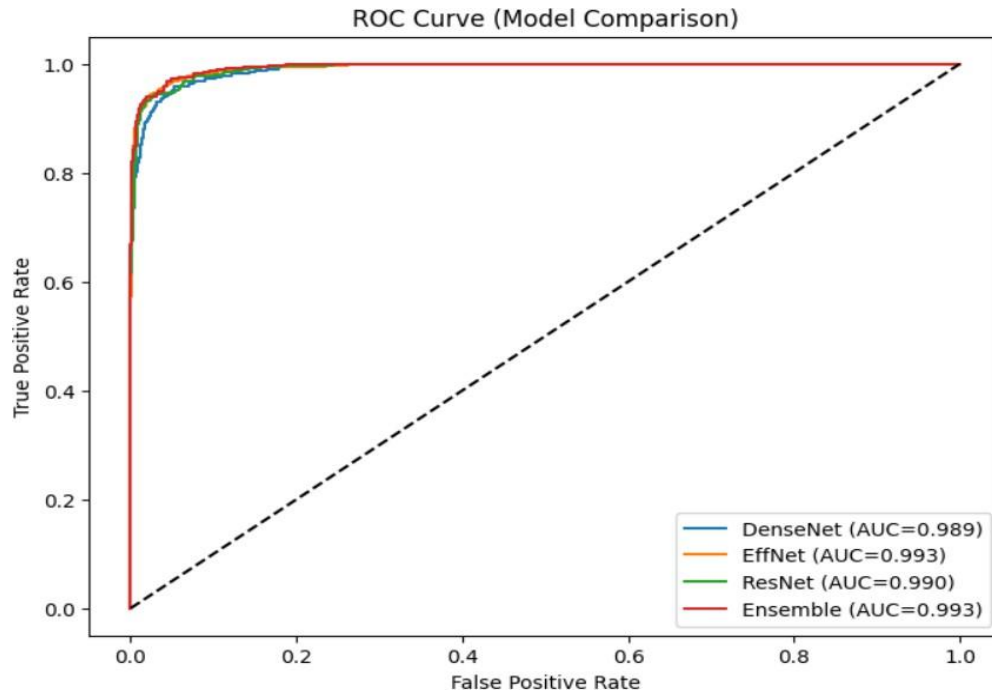


Figure 24: ROC Curve Analysis

The below Figure 25 shows Precision-Recall Curve.

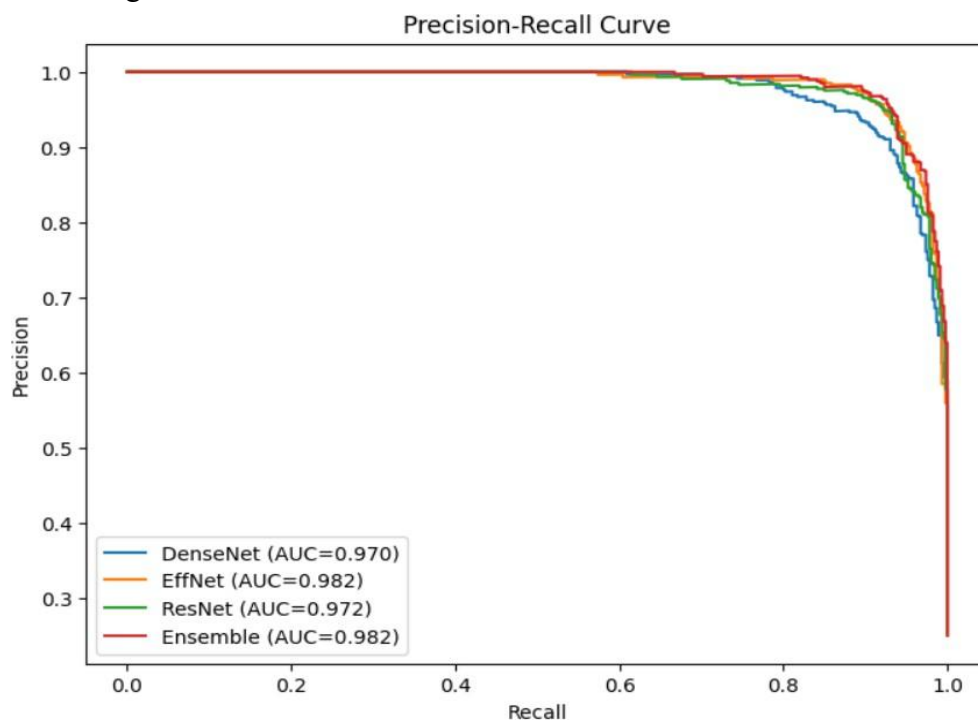


Figure 25: Precision-Recall Curve

7.5 Grad-CAM Visualization Results:

The below Figure 26 is Grad-CAM results highlight important regions in OCT images influencing the model's predictions. The heatmaps focus on disease-affected areas, confirming that the model learns relevant features. This improves interpretability and validates the model's reliability.

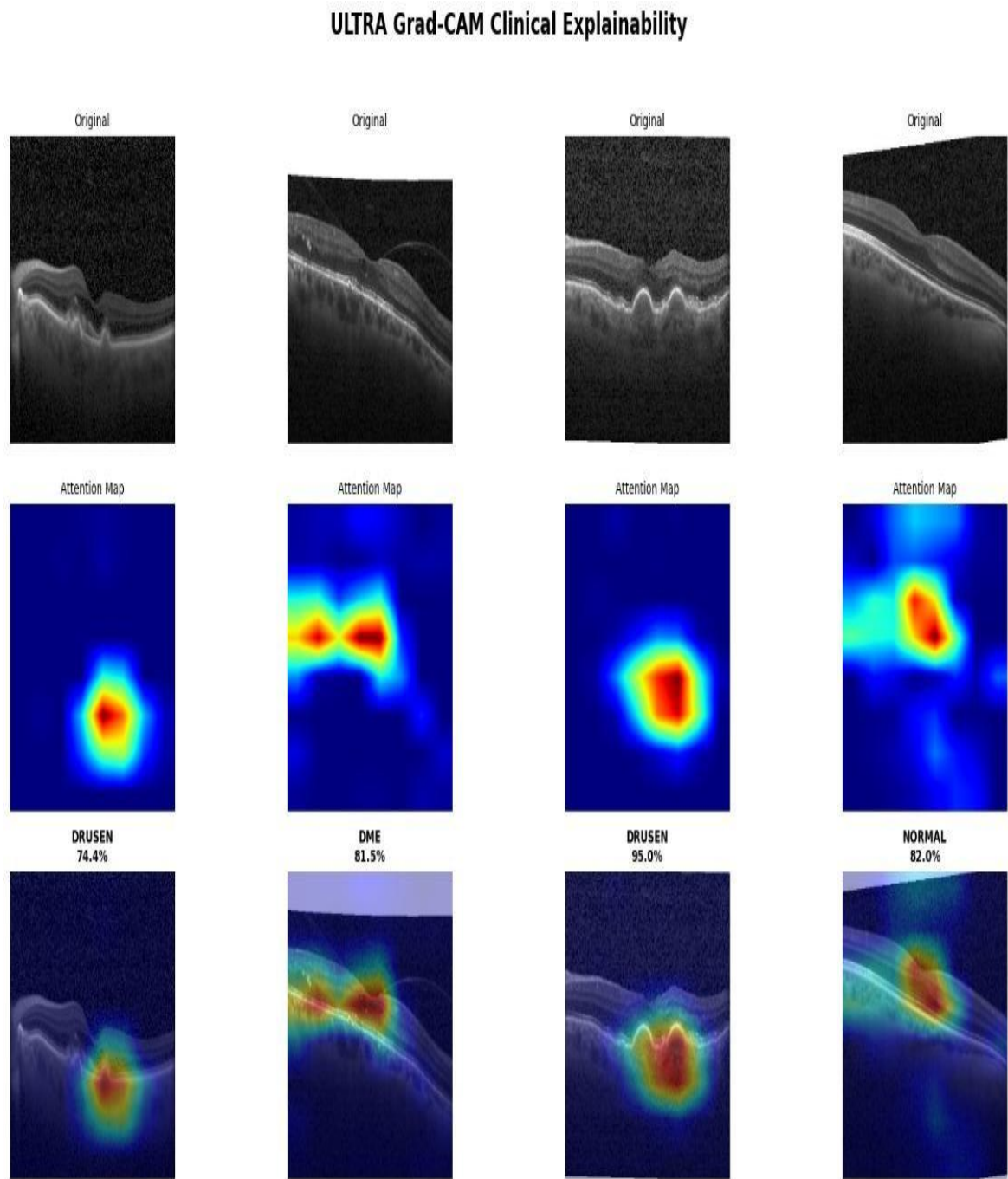


Figure 26: ULTRA Grad-CAM Clinical Explainability

7.6 Final Clinical Decision Outputs:

The results show that the system classifies OCT images and predicts disease severity with high confidence using ensemble learning. The clinical dashboard displays predicted classes with probability scores, while Grad-CAM highlights important regions influencing the predictions. This indicates accurate classification, reliable severity estimation, and clear interpretability of the model.

The below Figure 27 presents the clinical decision support dashboard.

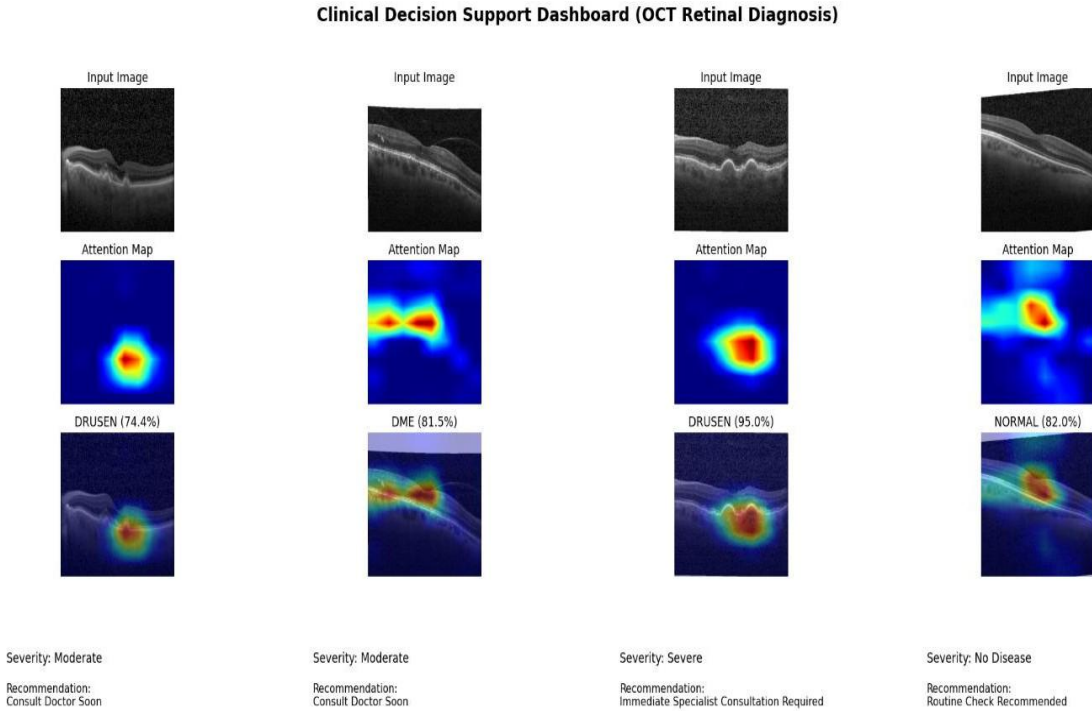


Figure 27: Clinical Decision Support Dashboard

The Figure 28 shows OCT multi-class severity prediction using ensemble learning.

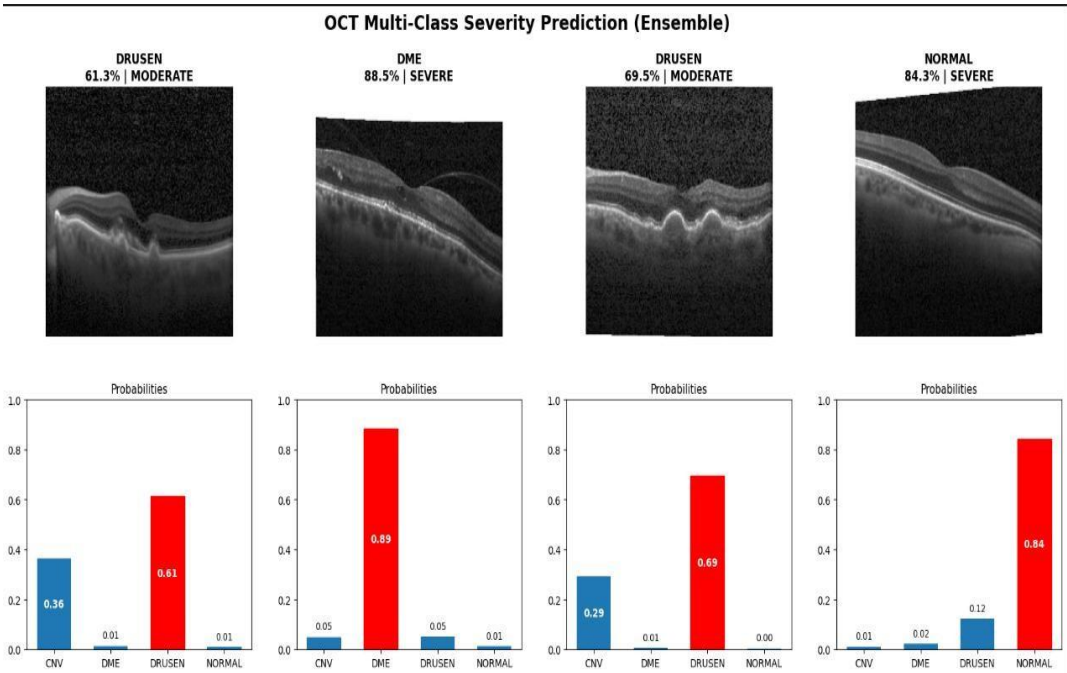


Figure 28: OCT Multi-Class Severity Prediction (Ensemble)

CHAPTER-8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion:

The proposed system for retinal disease classification using Optical Coherence Tomography (OCT) images demonstrates high accuracy and reliability through the integration of deep learning and ensemble learning techniques. Multiple pre-trained convolutional neural networks, including DenseNet121, EfficientNetB4, and ResNet50V2, were effectively utilized with transfer learning and fine-tuning to extract meaningful retinal features.

The ensemble model achieved a superior accuracy of approximately 94%, outperforming individual models and ensuring more robust and stable predictions. The use of advanced preprocessing and data augmentation techniques improved model generalization and reduced overfitting. Furthermore, evaluation using metrics such as precision, recall, F1-score, and confusion matrix confirmed balanced performance across all disease classes.

To enhance interpretability, Grad-CAM visualization was incorporated, providing clear insights into model decision-making by highlighting clinically relevant regions in OCT images. Additionally, the implementation of a multi-stage clinical decision pipeline enabled disease detection, classification, severity estimation, and confidence-based decision support, making the system more practical for real-world medical applications.

Overall, the proposed framework is efficient, accurate, and clinically relevant, demonstrating strong potential for assisting ophthalmologists in early detection and diagnosis of retinal diseases.

8.2 Future Scope:

The proposed system can be further enhanced by incorporating larger and more diverse OCT datasets to improve generalization across different populations and imaging conditions. Future work can also explore advanced deep learning architectures such as Vision Transformers and hybrid CNN-transformer models to achieve higher accuracy and better feature representation.

Real-time deployment of the system as a web or mobile-based clinical application can significantly improve accessibility for early diagnosis. Integration with hospital information systems and electronic health records can further support automated screening workflows. Additionally, extending the model to detect a wider range of retinal diseases and multi-modal data (such as fundus images) can improve diagnostic capability.

Further improvements can be made by enhancing explainability using advanced interpretability techniques and incorporating uncertainty estimation for safer clinical decision-making. Overall, the system can evolve into a comprehensive AI-powered diagnostic tool for ophthalmology.

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